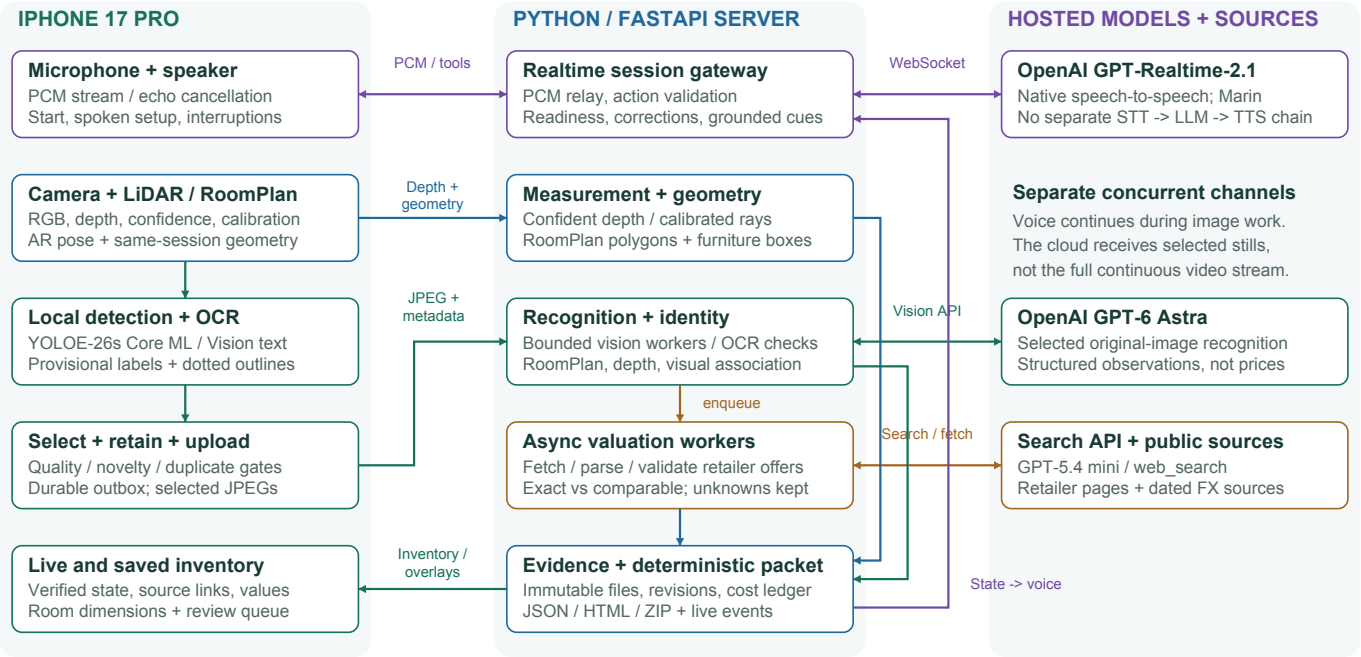
Library Contents Claim Agent - technical report



As-built snapshot: 20 October 2026. This report explains the native iPhone and paired Python service in this repository. It expands the one-page architecture note; the main diagram is also available as Mermaid source. Requirements come from the assessment brief (reviewer access required). Fresh software checks are recorded in submission validation.

The phone handles camera/depth acquisition, local proposals, OCR, preview tracking, capture selection and audio I/O. The Mac coordinates server vision, physical identity, deterministic geometry, sourced valuation, voice actions, persistence and packet generation. Models propose observations; validated server state controls claim numbers.

This is not a completed measured-accuracy submission. A saved physical sweep and repair evidence exist, but the required full library ground truth and pass-bar evaluation have not been established. The most recent pricing run encountered `credit_balance_exhausted`. Configuration and successful earlier provider probes do not establish present funded access.

The PDF includes a full landscape appendix with 12 numbered request/response flows and an endpoint index. The service flow guide provides individual SVG diagrams and editable Mermaid source covering setup, streamed audio, capture, recognition, LiDAR geometry, pricing, corrections, export, comparison, second-market valuation and recovery. Requests, responses, asynchronous events and internal calls are distinguished.

1. Execution boundaries and model responsibilities

Component	Runs where	Input and responsibility	Authority limit
ARKit and RoomPlan	iPhone	Shared RGB/depth/pose stream; surfaces, furniture boxes and capture coaching	Live measurements are provisional; accuracy needs physical validation
YOLOE-26s prompt-free INT8 Core ML	iPhone	640 x 640 inference on full image and two overlapping detail regions; complete 4,585-label vocabulary from metadata	Proposals, not accepted inventory, dimensions or prices
Apple Vision OCR	iPhone	Accurate text recognition; position/confidence mapped to calibrated image axes	Independent observed text, with possible omissions/errors
Apple Vision object tracking	iPhone	Carries dotted preview outlines between detections	Presentation only; does not create inventory or claim geometry
GPT-6 Astra	OpenAI, called by Mac	Structured inspection of upright full-resolution evidence: individual book spines, shelf levels and other objects	Output is validated; no generated dimensions, prices or totals
GPT-5.4 Mini web search	OpenAI, called by Mac	Discovers relevant product-page URLs	Generated prose is never price evidence
GPT-Realtime-2.1	OpenAI through Mac WebSocket	Streaming conversation, speech and guarded tool requests; low reasoning effort, `marin` voice	Reads server facts; validated tools control mutations
`gpt-4o-mini-transcribe`	Realtime input transcription	Asynchronous input transcript for display and action authorization	Ordinary output audio does not wait for transcription
Python / NumPy / Shapely / Decimal	Mac	Geometry, tracking rules, offer validation, revisions, totals, JSON/HTML/ZIP	Deterministic calculations and explicit rejection paths

The bundled model is approximately 12.5 MB. It is a compact vision detector, not a generative small language model. Its export preserves all 4,585 class IDs through an INT32-to-FP32 path; an all-FP16 cast had rounded IDs above 2,048. Provenance, hashes and the AGPL-3.0 or enterprise license are recorded in model documentation. Model confidence is an uncalibrated score, not measured accuracy.

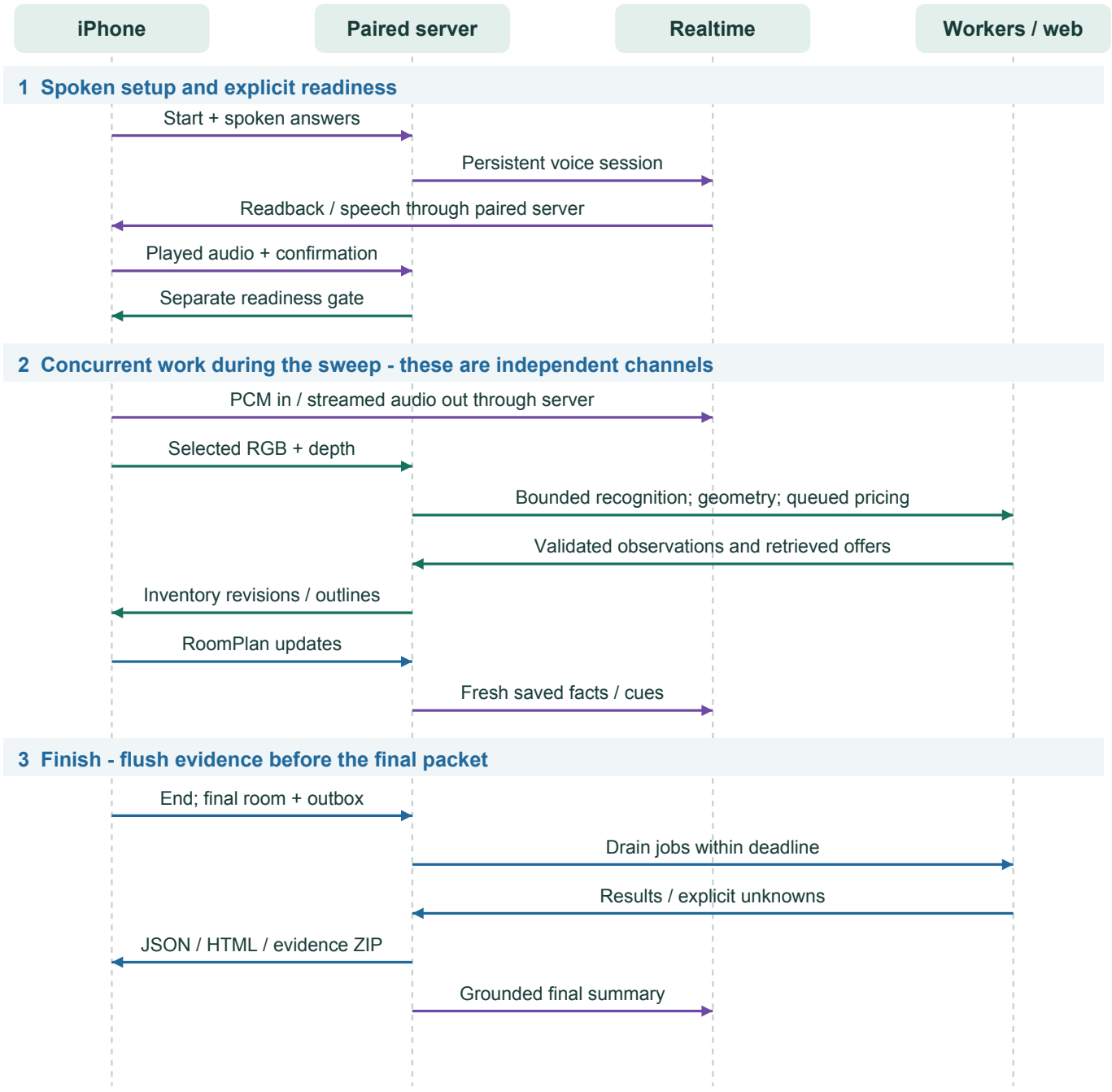
Active versus historical paths. Current `DeviceIntelligence.swift` returns `source=cloud_requested`, `status=unavailable`, empty semantic clues and `needs_cloud=true`. No Apple Foundation Models inference runs in this path. `local_first=true` means local proposals/OCR, capture quality, novelty and bounded escalation; it does not mean cloud-free semantic recognition. The server retains a validated local-clue/Open Library corroboration path for compatible inputs, but the current phone does not produce those semantic clues. Catalogue editions never become observed editions. Older descriptions of an on-device text model are historical.

Claude, ElevenLabs and Gemini adapters remain for earlier configurations and tests. Some shared classes retain historical names such as `ClaudeAssistant`. Active OpenAI routing uses Realtime for speech/conversation and Responses for vision/search; it does not silently fail over to those providers. Apple Speech recovery code is not the active Realtime voice route.

2. One sweep: timing and coordination

The camera preview may start before setup, but evidence saving and the sweep clock wait for readiness. Spoken setup collects language and a supported market: India/INR, United Kingdom/GBP or United States/USD. Sensor capability selects LiDAR where available; camera-only is the fallback on supported non-LiDAR hardware. A temporary depth gap never silently changes a confirmed LiDAR session into camera-only capture.

The setup gate associates the complete readback with its Realtime response and the phone's actual playback acknowledgment. A distinct intelligible owner turn confirms setup; a later distinct turn answers readiness. Interrupted prompts, noise, missing transcripts and stale responses cannot satisfy these gates. The simulator supports conversation checks but cannot supply real capture evidence.



Preview video stays on device. Only selected stills and calibrated sensor evidence are uploaded.

These branches have separate queues and lifecycles. Recognition and pricing need not finish before the user hears a reply. The active upload path supplies structured observations to voice rather than a second billable copy of every JPEG. Although the shared bridge supports image messages, routine sweep uploads do not call that path.

Audio uses AVAudioEngine voice processing, a dedicated dispatch queue, 16-bit mono 16 kHz input, stateful server resampling to 24 kHz and streamed 24 kHz output. Semantic VAD supports automatic barge-in; **Interrupt & speak** explicitly cancels playback. The server truncates conversation audio to the acknowledged played duration. Generation completion alone does not mean the owner heard it. Backgrounding, route interruptions, stop and reconnect invalidate stale callbacks and queued audio. Passive silence does not request periodic paid check-ins.

The cue ledger distinguishes upload accepted, recognition processed, existing physical records, missing clarity, sourced value and final processing. Before speech, it rechecks the current view and inventory revision. Duplicate image receipt alone never proves that an object was counted, and move-on guidance requires usable saved evidence. Owner

corrections target exact IDs; ambiguous references require clarification. Claims about a first edition are stored as owner statements for review, not promoted to verified visual facts.

3. Phone capture, overlays and evidence

One ARSession supplies camera pixels, intrinsics, pose and LiDAR coordinates; RoomCaptureSession uses that session. Capture chooses the highest-resolution supported camera format at no more than 30 fps. The preview recognition timer requests work every 0.25 seconds, but in-flight work is guarded; this is not a claimed four-inference-per-second benchmark.

The detector runs a full-frame view and two overlapping detail crops. All native classes are eligible. Generic geometry maps proposals back to the original calibrated image, rejects crop-edge truncation and preserves full-image labels over conflicting detail predictions. At most 300 proposals enter metadata. There is no room-specific class list or hardcoded catalogue.

Capture requires normal AR tracking, adequate sharpness/exposure, fewer than four pending uploads and a minimum interval: four seconds with detections or ten seconds for an otherwise clear overview. Conservative visual-hash and camera-motion checks suppress near repeats. Improved sharpness or depth readiness can admit another view. Clear changed-scene overviews remain eligible even if the detector misses a class. Clarity requests are throttled to avoid repeatedly interrupting the owner.

The current UI draws dotted object borders with labels and available measurements. It does not display filled segmentation masks or projected mask textures; retained mask decoder fixtures support model validation/history. Local screen tracking uses resized frames and a bounded short in-memory history to handle detector delay; it expires lost/stale tracks. Those thumbnails are not evidence and are not uploaded. Measured world overlays have a separate supported-geometry path. A border on screen is not a counted or priced claim line.

`FrameEncoder` retains original-size JPEG evidence at quality 0.92, unrotated image dimensions, normalized polygons, column-major camera matrices, capture-relative time and AR-session ID. Depth is aligned little-endian float32 metres; confidence is one byte per depth sample. Server vision sees an upright image and returned polygons are mapped back before geometry. Camera-only serialization removes depth/confidence; the server rejects either depth or RoomPlan geometry in that mode.

The disk outbox writes each complete payload atomically before upload. A frame ID cannot later contain different bytes. Pending uploads survive connection failures and retry with backpressure delays; acknowledgments move files to retained uploaded storage. The outbox has a 1 GB storage bound. Preserving payloads does not establish automatic restoration of a live conversation or continuous AR coordinates after app termination.

4. Metric scale and geometry mathematics

Scale comes from LiDAR/ARKit metric observations, not a model's knowledge of a typical book or room. A four-corner spine polygon alone has no physical scale.

For pixel `(u,v)`, focal lengths `(fx,fy)` and principal point `(cx,cy)`, the camera ray in the implementation's ARKit convention is:

```
r(u,v) = [(u-cx)/fx, -(v-cy)/fy, -1]
p(u,v,z) = z r(u,v)                # measured forward depth z, metres
n · (p-c) = 0                      # fitted plane, center c, normal n
t = (n · c) / (n · r)
q = t r                            # corner/plane intersection
q_world = T_camera_to_world [q,1]
```

Intrinsics and transforms are reconstructed from column-major arrays. Normalized polygon corners become raw JPEG pixels. The server expands the polygon's depth neighborhood by four depth pixels, keeps high-confidence depths between 0.15 and 5 m, retains samples within 8 cm of the median and requires at least 16 samples. SVD fits a plane; residual RMS above 1.5 cm, insufficient spread or an oblique ray denominator below 0.15 rejects the result. The four ray intersections yield edge lengths. The two opposite-edge averages become the longer spine side and shorter thickness, multiplied by 100 for centimetres. Implausible sizes and invalid calibration return `null` plus a review reason. These thresholds are engineering gates, not a calibrated uncertainty model.

For non-book objects, a planar visible face can provide two face-edge measurements. Those are not a full three-dimensional bounding box. Full width/height/depth comes from valid medium/high-confidence RoomPlan furniture

boxes, linked conservatively to a visual item. Revised geometry clears obsolete dimensions before reassociation. Robust depth support used for tracking is identity-only evidence and never becomes object width or depth.

Floor area is the union of captured horizontal XZ polygons, preserving concavity. Length/width are the longer/shorter sides of the minimum rotated bounding rectangle; their product is not substituted for polygon floor area. Wall area sums valid planar wall polygons. Height is the median wall height, with variation over 10 cm flagged. Openings and shelving are projected into each associated wall plane, unioned and clipped to the wall. Gross wall, openings, net wall and shelving coverage are separate outputs. Missing/ambiguous shelving association leaves coverage unknown. Curved surfaces are flagged rather than assigned a planar substitute. Square feet use `m2 x 10.76391041671`.

Occupied shelf run sums unioned horizontal book-footprint intervals within each supported shelf level, avoiding double-counting stacks. Missing required book/shelf geometry leaves the run unknown. Camera-only mode has no implemented monocular reconstruction, reference-object scale or reliable general metric association; room areas and dimensions remain unknown.

5. Recognition and three duplicate safeguards

Server admission validates JPEG type/calibrated dimensions, depth/confidence byte lengths, session continuity and immutable signatures. Quality checks run independently of phone claims. In active local-first mode, clear novel unresolved views may reserve a vision attempt; the reservation is saved before the provider call. Low-quality views, exhausted budgets, cooldowns and already attempted evidence remain explicit routing outcomes.

GPT-6 Astra receives the whole image and untrusted local hints. Its structured result separates individual book spines, shelf levels and other objects. Each quadrilateral is validated independently; one malformed polygon does not discard all valid detections. Books require visible text supported by independent OCR; unreadable titles stay blank and unidentified. Edition and ISBN guesses are not filled from catalogue familiarity. Model omission does not erase previously supported local identities.

• **Image duplication:** SHA-256 identifies exact JPEG bytes; perceptual similarity additionally requires a nearly stationary pose. Content-addressed JPEG blobs are hard-linked to per-frame references. Exact repeats retain their own metadata, depth and confidence. Existing RGB polygons may be remeasured against newly useful depth without a second vision call. A different pose cannot reuse a dimension just because JPEG bytes match. Near-repeat frames carrying depth are retained conservatively.

• **Physical identity:** `TrackRegistry` checks kind, compatible category, AR session and one-to-one frame membership. It prefers a matching projected RoomPlan object, then reliable observed depth support or measured corners. For supported non-book cases it can use calibrated visual overlap and bounded SIFT scene correspondence with local geometric support. Contradictory measured depth blocks a visual merge. Competing candidates are rejected as ambiguous; name equality never merges physical copies. Tracks retain at most eight views. This reduces duplication but does not prove all physical counts.

• **Repeated work:** Vision reservations and signatures survive failures. Pricing has durable status, record revision and query fingerprints; equivalent concurrent lookups are coalesced. Stale/interrupted work cannot silently restart a paid request or overwrite a correction. Provider response IDs suppress duplicate usage accounting.

The conservative repair function replays hash-verified observations in a read-only preview; it neither calls providers nor saves by itself. It keeps uncertain mappings, original observation IDs, aliases and prior invalidated prices auditable. The recorded application to a saved 21-frame sweep reduced **87 to 76 item records**, including **four chair records to one**, with three books unchanged. Four ambiguous identities and one related target were untouched; original state was backed up. This demonstrates repair of saved evidence, not a ground-truth assertion of 76 physical items or a new full library capture.

6. Sourced prices, revisions and output

Identification, measurement and pricing are separate stages. Pricing admits identified eligible records only. Books request new replacement and used value separately; used price is never inferred by discounting new price. Other objects request replacement offers using an exact supported product or an explicitly labelled equivalent/range when sufficient attributes support it. Art/portraits default to appraisal until explicit owner print confirmation; rare/signed signals and retrieved values above INR 10,000 equivalent also require appraisal.

Search returns at most eight public product URLs from citations/source metadata. The server independently fetches pages and parses structured JSON-LD products/offers. Acceptance checks product/variant identity, observed brand/model or

supported equivalent, material/dimensions where available, condition evidence, stock, currency and market. Search answer prose never supplies a price. No retrievable supported offer means unknown and exclusion from totals. Blocked pages and unstructured retailers remain a coverage limitation.

Each accepted price preserves amount/range, market/currency, condition assumption, exact/equivalent match basis, product/source URL, retrieval timestamp and saved extraction reference. Search traces are retained. If a local offer is unavailable, foreign fallback searches the UK, or India for a GBP target, then converts using retrieved Frankfurter FX evidence. Original amount/currency, rate, FX URL/date and a converted flag remain visible. No current exchange rate is hardcoded.

Price and URL-search caches are separately bounded to 256 entries with a 900-second TTL. Equivalent concurrent queries share in-flight work. Search terms can be reused when dimensions change, but each actual dimensional query still validates offers. Revision/fingerprint checks prevent a quote for an old identity, ownership state or measurement entering the current record. Corrections invalidate dependent valuations. Earlier provider work with ambiguous attribution is not silently replayed when opening saved inventory.

The same ten identified physical book IDs can be requested in a second supported market without changing the primary locale. This capability is not evidence that the required ten-book comparison has been completed.

`packet.py` builds typed `claim_packet.json`, derives `report.html` from the same serialization and packages immutable evidence into `packet.zip`. Monetary arithmetic uses `Decimal`; unknown numbers are `null`, and money serializes as decimal strings. Every book/item frame reference must resolve. Prices lacking eligible identity, dated source, condition or conversion evidence are removed from totals. Review entries name the affected record and reason. Completeness is `partial` when review needs remain. Runtime code never reads ground truth.

7. Runtime settings, costs and failure behavior

This snapshot separates verified active settings from source defaults/limits. It contains no secrets.

Control	Active setting or implemented bound	Consequence
Provider / vision / voice	OpenAI / `gpt-6-astra` / `gpt-realtime-2.1`	Historical providers inactive
Local-first / paid pricing / cloud voice	All enabled	Selected images and eligible valuation may incur cost
Vision routing	20 attempts/sweep; minimum 6 s apart	Source defaults are 3 attempts / 30 s; workspace permits more
Frame workers / server queue	2 workers / 12 queued frames	Backpressure keeps phone evidence on disk
Pricing workers / queue	4 workers / 250 jobs	Independent of audio and frame queue
Provider adapter cap	240 requests per provider instance	Separate from 20-image admission; vision/search share adapter
RoomPlan upload	At most every 3 s; one in flight	Coalesces provisional updates; finalization awaits current work
Frame/body bounds	360 frames/sweep; JPEG <= 10 MB; HTTP body <= 22 MB	Oversized/excessive evidence rejected
Final processing wait	Source default 240 s, configurable ceiling 270 s	Partial results on deadline; no guarantee of total under-five-minute acceptance
Quote deadline	45 s per quote; four concurrent	Timeout leaves unknown; no automatic paid replay
Appraisal threshold	INR 10,000 equivalent	High-value offers retained as evidence, excluded from automatic value

Live audio has a separate bounded queue. Dropped audio chunks are counted and trigger a request to repeat missed speech. The event stream publishes revisions and overlays independently of voice generation. This is a local

single-process demonstration with bounded concurrent sweeps, in-process workers and filesystem snapshots, not a distributed durable job system.

The cost ledger uses reported usage, including cached input, output and Realtime text/audio/image modality splits, input transcription and reported search-tool calls. Its dated rate card is in `costing.py`. In general, cost is `(uncached input x input rate + cached input x cache rate + output x output rate) / 1,000,000`, summed by model/modality, plus supported search/cache-write fees. It counts reported tool calls rather than assuming one execution per API request. Onboarding usage transfers into the sweep once. Estimates are not invoices and exclude hosting, tax, device and network expense. Unknown usage/rates and ambiguous interrupted requests are uncosted; a partial known subtotal is not the full bill.

Telemetry includes `identification_stages`, `pricing_stages`, `provider_usage`, `live_usage`, `live_first_audio_ms`, `backend_finalize_s` and `time_to_packet_s`. Final timing conservatively includes native finalization/upload delay using server creation plus native monotonic capture duration. Provider first-audio timing differs from phone audible latency. No end-to-end latency or accuracy is inferred from health or a synthetic fixture.

Failure	Implemented response
LAN loss or backpressure	Retain atomic phone payloads; retry evidence upload with delay
Blur, darkness, glare or crop	Request improved view within the continuous pass; do not invent hidden objects
Invalid depth/calibration	Preserve visual evidence; unknown measurements and review reasons
Billing/access error	Latch further affected provider work off; retain unknowns and error usage
Interrupted/uncertain paid request	Mark uncosted/unknown; do not automatically repeat
Late quote after correction	Revision and eligibility checks reject stale results
Final processing deadline	Cancel unfinished workers, exclude unfinished values, emit partial packet
Server restart during capture	Mark partial; require a new continuous capture rather than join AR sessions
Audio stop/background/route failure	End/reconnect voice with stale callbacks discarded; visuals have their own lifecycle

8. Module map

Area	Main files	Review focus
Native orchestration/transport	`CaptureController.swift`, `BackendClient.swift`, `EvidenceOutbox.swift`	Session lifecycle, setup, durable evidence and finalization
Native recognition/selection	`LocalObjectDetector.swift`, `LocalRecognitionModels.swift`, `FrameEncoder.swift`, `DeviceIntelligence.swift`	Proposals, OCR, coordinates and cloud request
Native display tracking	`LocalPreviewTracker.swift`, `PreviewTrackingModels.swift`, `ObjectOverlay.swift`, `OverlayModels.swift`	Dotted outlines, freshness, separation from inventory
Native measurements/audio	`RoomGeometry.swift`, `MeasurementModels.swift`, `LiveAudio.swift`	Geometry encoding/coalescing and playback accounting
API/setup	`server.py`, `config.py`, `onboarding.py`	Pairing, bounds, provider routing, spoken authorization
Sweep coordination	`pipeline.py`, `models.py`	Independent queues, revisions, corrections, final deadline
Recognition/routing	`vision.py`, `local_evidence.py`, `image_selection.py`, `openai_provider.py`	Structured output, OCR, novelty, spend reservation
Identity/scale	`tracking.py`, `object_anchor.py`, `visual_correspondence.py`, `geometry.py`	One-to-one matches, conflicting evidence, dimensions
Live interaction	`openai_live.py`, `live.py`, `sweep_guide.py`	PCM, playback gates, tools and current guidance
Valuation/evidence	`pricing.py`, `providers.py`, `storage.py`, `packet.py`, `costing.py`	Retrieved offers, hashes, arithmetic, usage provenance
Replay/evaluation	`inventory_repair.py`, `comparison.py`, `evaluate.py`, `evaluation/compare_packets.py`	Audit-preserving repair, depth ablation, external ground truth

Native paths are under `ios/LibraryClaim/`; unprefixed server files are under `claim_agent/`. Tests under `tests/` and `ios/Tests/` cover geometry, identity, price revalidation, voice gates, reconnects, outbox and overlays.

9. Setup, privacy and security

Development uses a Mac with Python 3.12, pinned `requirements.lock`, Xcode/signing and a physical ARKit-capable iPhone. LiDAR measurement requires supported hardware; this workspace uses an iPhone 17 Pro. The server starts with `python -m claim_agent` on port 4180. Follow repository setup and physical validation. Toolchain installation, signing and account provisioning are prerequisites, not part of a measured fifteen-minute fresh setup claim.

Credentials remain in ignored server configuration. `scripts/configure_ios.py` copies only local host, port and pairing token into ignored phone build configuration. A stable `.local` hostname avoids Wi-Fi IP churn. Wildcard binding uses IPv4 and IPv6 where supported; default loopback binding must be changed for phone access. `/api/health` reports configuration, not secrets or proven credit.

HTTP endpoints and WebSockets enforce shared pairing; same-origin checks, payload limits, path validation and hashes protect the evidence interface. Source retrieval allows public HTTP(S) on standard ports, validates redirects, rejects local/private destinations, bounds downloads and never forwards provider credentials. Image text and pages are data, not instructions authorizing mutations. HTML rendering escapes content.

This is demo security: LAN HTTP/WebSocket traffic is unencrypted, and the shared pairing secret grants access. Use a trusted network; production would require TLS, individual authentication/authorization and retention controls. Frames, OCR, transcripts/events, geometry and receipts may contain sensitive household information. Selected images go to OpenAI vision; microphone audio streams to OpenAI Realtime. The application does not persist raw microphone PCM, but retains transcripts/events and evidence locally. Responses requests set `store=False`; that does not establish a blanket provider-retention guarantee. Automatic face/document redaction and scheduled evidence deletion are not implemented. The operator must avoid people/private documents, obtain permission and manage retained files.

10. Evidence, limitations and assessment coverage

Latest recorded software verification is **362 passing Python tests**, scoped repair regressions and a clean whitespace/diff check. Native signed builds, installation, small real-photo checks and voice troubleshooting are documented historically. These prove execution of specific components, not whole-room accuracy. The small detector timing sample must not be generalized to continuous phone throughput.

The saved 21-frame sweep demonstrates evidence persistence and the reported duplicate repair. Its three recorded books and 76 unvalidated item records do not satisfy the required >= 60-book, two-shelving-unit, >= 8-non-book-item evaluation dataset. Floor area and room length/width are unknown in this saved result. Remaining ambiguous mappings are exposed. A sourced chair comparable exists in the repair history; it does not establish room-wide retailer coverage. Later pricing exhausted credit and left unsupported values unknown.

The saved usage ledger reports a **known cost subtotal of USD 6.01116405, with partial cost status: 191 costed and one uncosted request**. This is the implemented API-usage estimate, not a complete billed sweep cost. It must not be rounded into a claim that all provider charges are known.

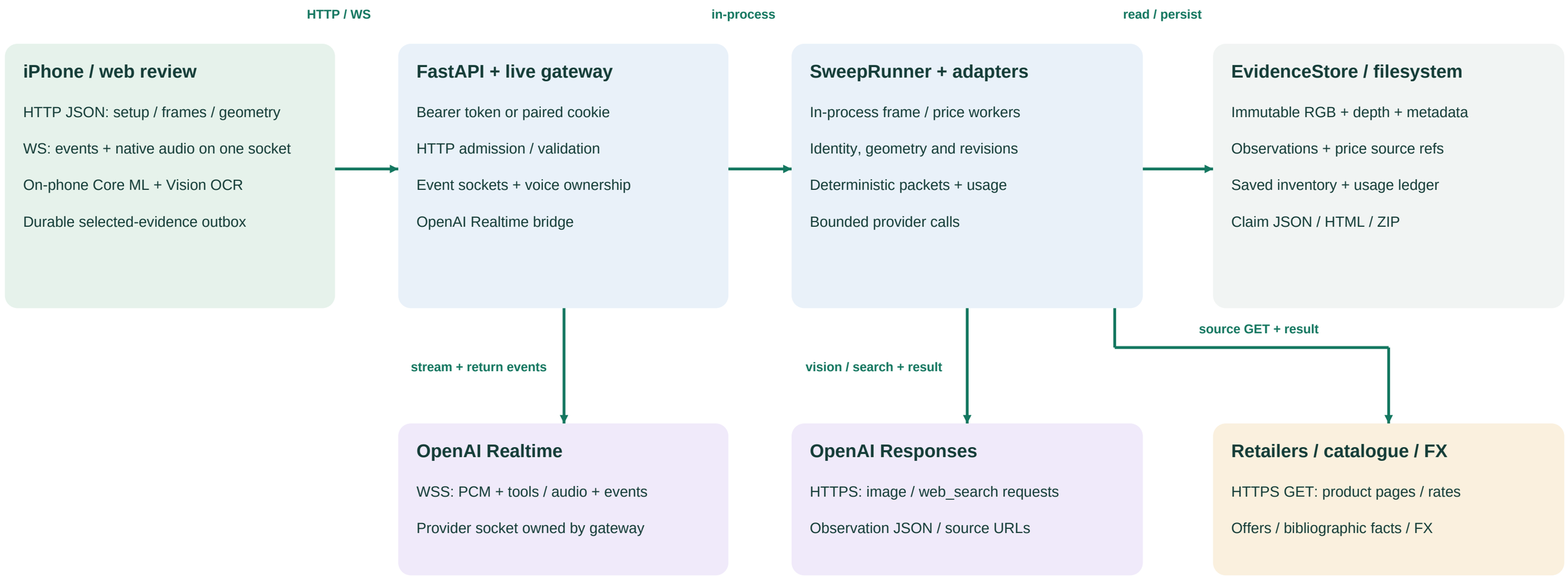
Assessment measure	Required bar	Supported status
Book count	Within 5% of true count	Not established against full physical ground truth
Title identification	>= 70% of human-legible spines; <= 3% confidently wrong	Not established
Spine dimensions	Within 15% on 20 measured spines	Method/regressions exist; physical sample not established
Book prices	Within 25% on 15 hand-checked prices, same source type	Validation pipeline exists; required sample not established
Non-book detection	>= 80%, correct category	Repair is not a complete labelled detection evaluation
Floor / wall area	Within 10% / 15% of tape measurements	Method exists; physical comparison not established
Packet latency	Under 5 minutes after sweep	Bounds/telemetry exist; scored run not established
Second locale	Same 10 books	Workflow exists; completed comparison not established

Key limitations are thin/reflective or occluded spines, broad detector false positives, nonplanar objects, incomplete RoomPlan surfaces, cautious merging and retailer JSON-LD/access coverage. Confidence scores are uncalibrated. Quality warnings can remain in review until a supported resolution exists. Masks are not a current display feature. There is no continuous video-to-model stream: continuous camera preview yields selected still-image evidence while audio streams separately.

The optional camera comparison freezes a LiDAR result and independently reanalyses its JPEGs without depth, RoomPlan, inherited inventory or prices. It preserves hashes and independent usage. Reused AR poses may have benefited from LiDAR: this is same-frame depth ablation, not a pure hardware benchmark. Replay latency differs from capture latency; response deltas need independent physical-ID mappings and frozen ground truth before they become accuracy results.

Request / response architecture

One native client, one FastAPI process, external model and source APIs. Internal modules are not separate deployed microservices.



Routes show the caller. Reverse responses/events use the same connection; the numbered sequences below show each direction separately.

Detailed sequence index

F01 Open a session and connect voice

F02 Spoken setup, readiness and capture handoff

F03 Stream audio, acknowledge playback and interrupt

F04 Select, retain and upload camera evidence

F05 Recognize a frame and publish inventory

F06 Measure objects and update RoomPlan geometry

F07 Search, verify and apply a price

F08 Read inventory and apply corrections

F09 Finish, generate and download claim evidence

F10 Compare camera-only with LiDAR

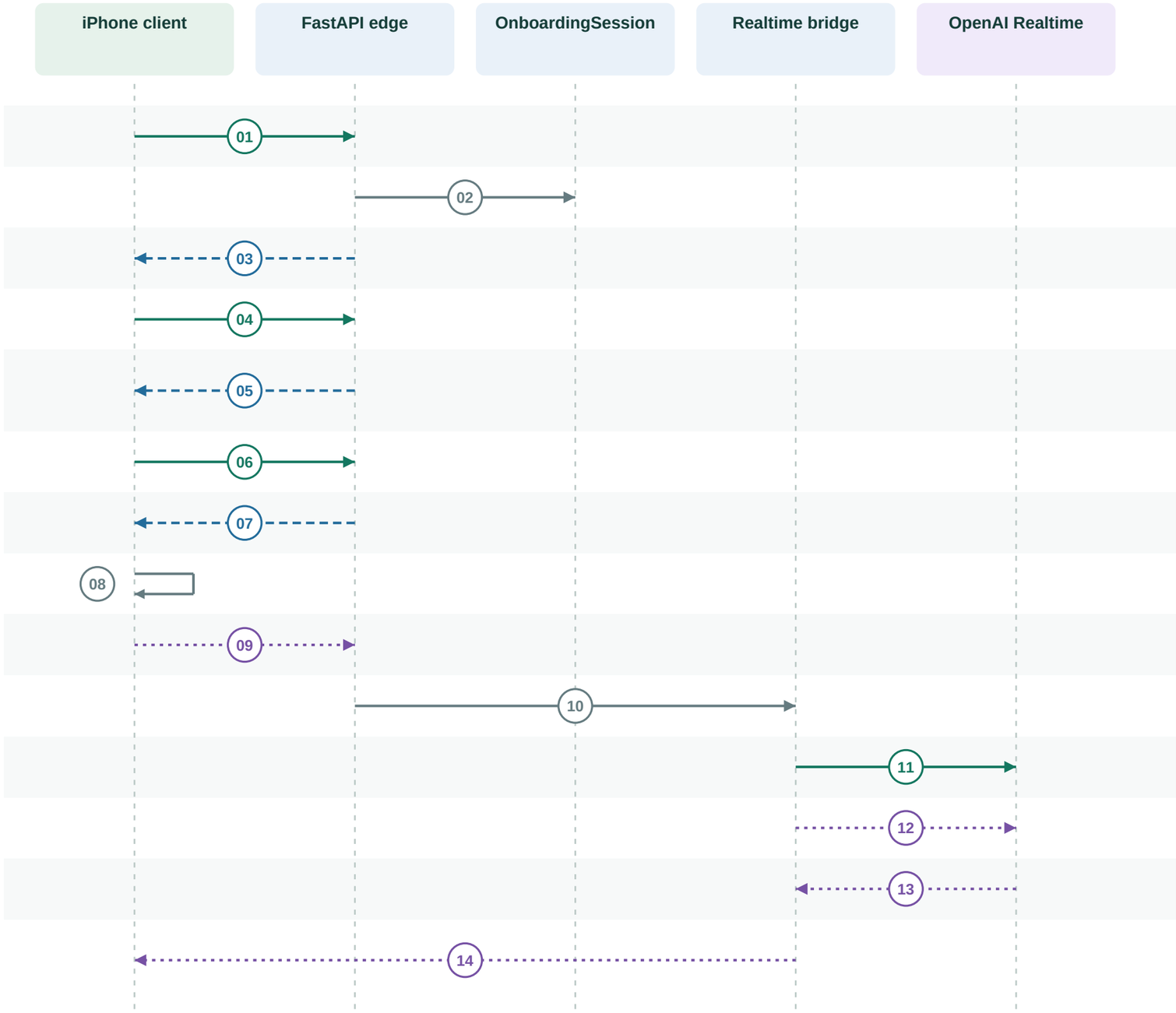
F11 Reprice the same books in a second market

F12 Reconnect, recover and account for usage

Green solid: network request | Blue dashed: network/result response | Purple dotted: asynchronous event | Grey solid: internal call

F01 Open a session and connect voice

HTTP creates setup state; the client WebSocket and provider WebSocket are separate connections.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 POST /api/onboarding
JSON: device, capture_supported (LiDAR), camera_supported.

02 Create bounded in-memory setup session
Expire/consume old sessions; at most 8 live setup sessions; TTL 600 s.

03 200 {id, expires_in_s: 600}
Alternative: 409 conversation limit; common auth/schema errors apply.

04 GET /api/health
Read advertised OpenAI voice transport/readiness; no provider request.

05 200 capability/configuration JSON
Require voice_transport=realtime, assistant_provider=openai, realtime_ready=true.

06 WS /api/onboarding/{id}/live
Authenticated WebSocket handshake; startVoice:false in native client.

07 WebSocket accepted; listener attached
Independent event sender; stored setup events may be replayed.

08 Prepare AVAudioEngine and event tasks
Settle voice-processing audio graph before asking for paid voice.

09 start_voice
JSON control event; await voice_ready or voice_error, not an HTTP reply.

10 Select OpenAILiveBridge and claim voice owner
A second voice owner gets voice_error; event socket can remain open.

11 WSS api.openai.com/v1/realtime?model=...
Server credential; configured gpt-realtime-2.1 in submission configuration.

12 session.update
24 kHz PCM, Marin, semantic VAD, tools, asynchronous input transcription.

13 session.updated
Provider acknowledges configuration; wait bounded to 20 s.

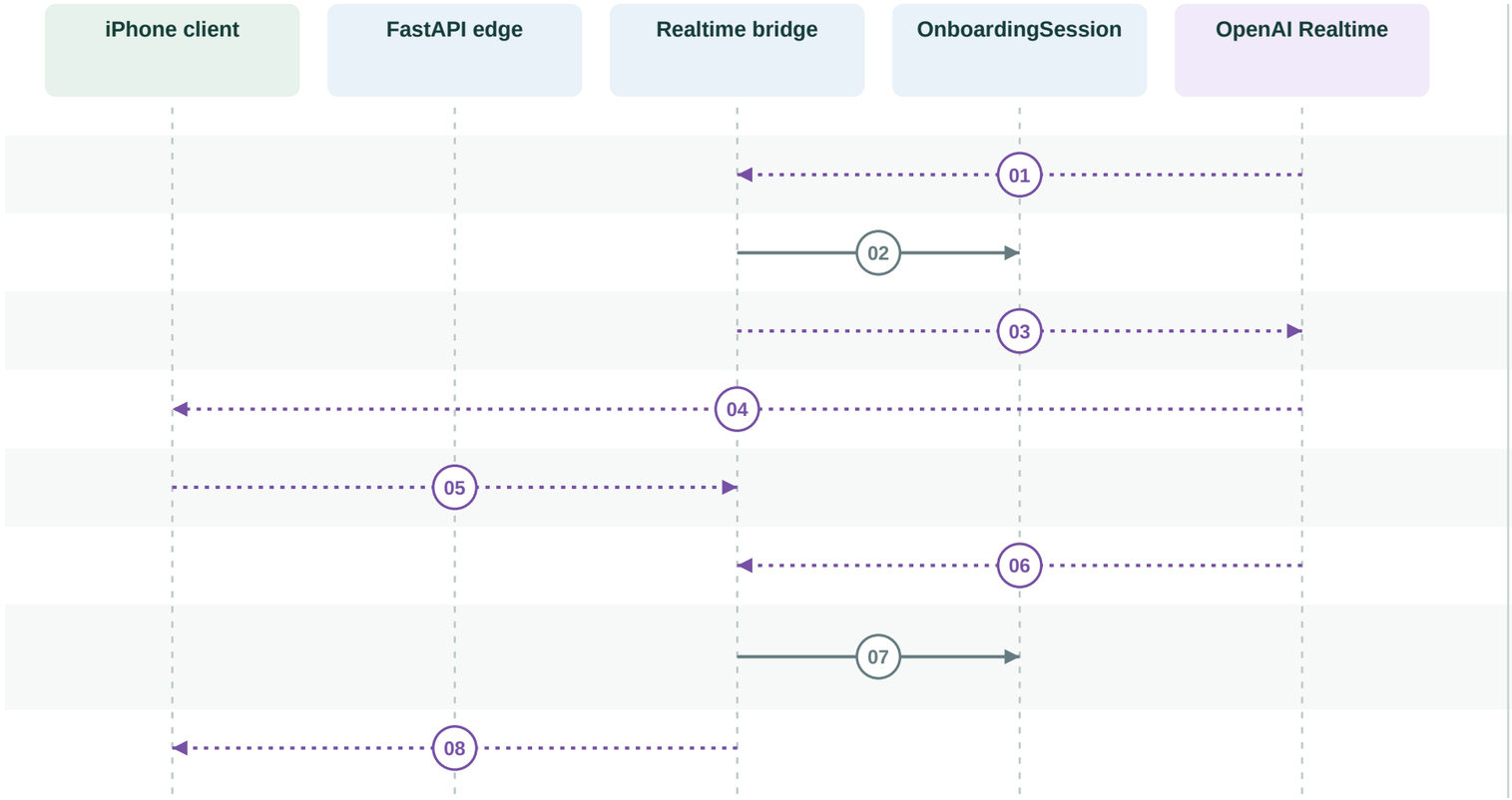
14 voice_ready (via FastAPI event sender)
Server enqueues voice_ready and requests the greeting; native enables microphone when voice_ready arrives.

BOUNDARY The native app sends a bearer pairing token; it does not call /api/pair. Health reports configuration, not funded access. A voice_ready event, not socket open, enables microphone sending.

SOURCE claim_agent/server.py:262 | claim_agent/server.py:422 | claim_agent/openai_live.py:633 | ios/LibraryClaim/CaptureController.swift:584

F02 Spoken setup, readiness and capture handoff - part 1/2

Tool requests are provider events; the server validates them and changes setup state in process.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 **response.function_call_arguments.done**
present_setup: language, country, currency, capture_mode, prompt templates.

02 **Validate presented preferences and owner turn**
Render readback server-side; select LiDAR when supported, otherwise camera.

03 **conversation.item.create + response.create**
Function output by call_id; spoken full readback requested with tools disabled.

04 **Assistant audio via bridge and API**
response.output_audio.delta -> audio; transcript/done tracked separately.

05 **playback via API: actual played_ms**
Audio completion + output transcript + played coverage open confirmation gate.

06 **Later owner turn: configure_session**
Matching preferences with confirmed:true after delivered readback.

07 **Store confirmed preferences in memory**
Onboarding session only; emit setup_confirmed and schedule separate readiness question.

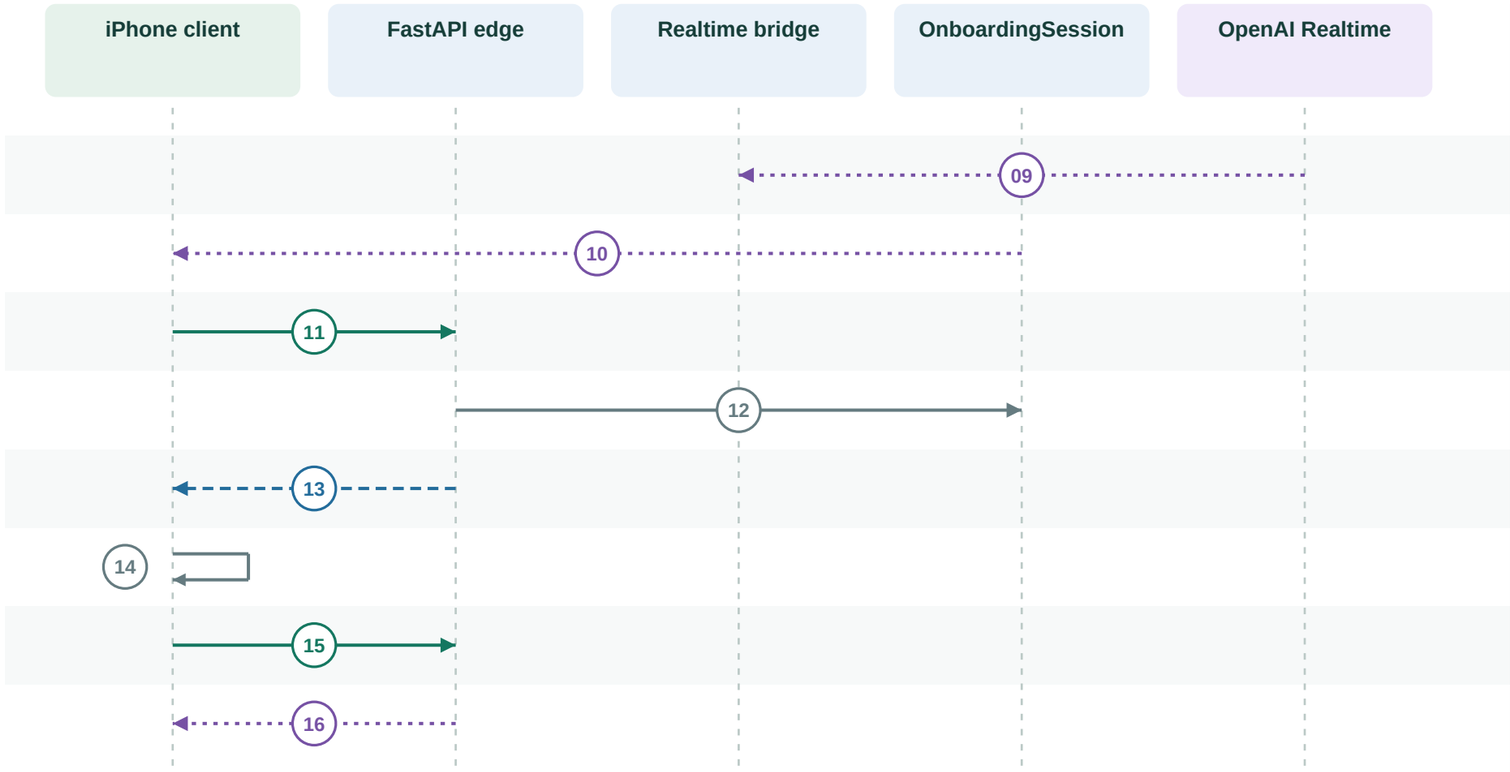
08 **setup_confirmed {preferences}; readiness audio**
Phone acknowledges actual readiness-question playback before next gate.

BOUNDARY Confirmation intent is model-interpreted. Code enforces later intelligible owner turns, gate versions and matching arguments; playback must cover readback and readiness. Preview may run, evidence capture does not.

SOURCE claim_agent/openai_live.py:187 | claim_agent/openai_live.py:426 | claim_agent/onboarding.py:110 | claim_agent/server.py:211 | ios/LibraryClaim/CaptureController.swift:205

F02 Spoken setup, readiness and capture handoff - part 2/2

Tool requests are provider events; the server validates them and changes setup state in process.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

09 **Later owner turn: request_capture {ready:true}**
Validate readiness phase, intelligible turn and physical sensor capability.

10 **capture_requested via API**
Native closes onboarding voice/events before creating the sweep.

11 **POST /api/sweeps**
JSON: device, onboarding_id. Retry lookup first checks persisted handoff.

12 **Consume setup and copy preferences / usage**
Persist onboarding_id with new sweep; close setup voice, refresh copied usage.

13 **200 ClaimPacket snapshot**
Same onboarding ID recovers existing sweep; invalid/expired handoff is 409.

14 **Start shared AR/RoomPlan session**
Create capture outbox and clock; choose mode from returned sweep state.

15 **WS /api/sweeps/{id}/live**
New sweep event connection; a new start_voice follows F01 provider setup.

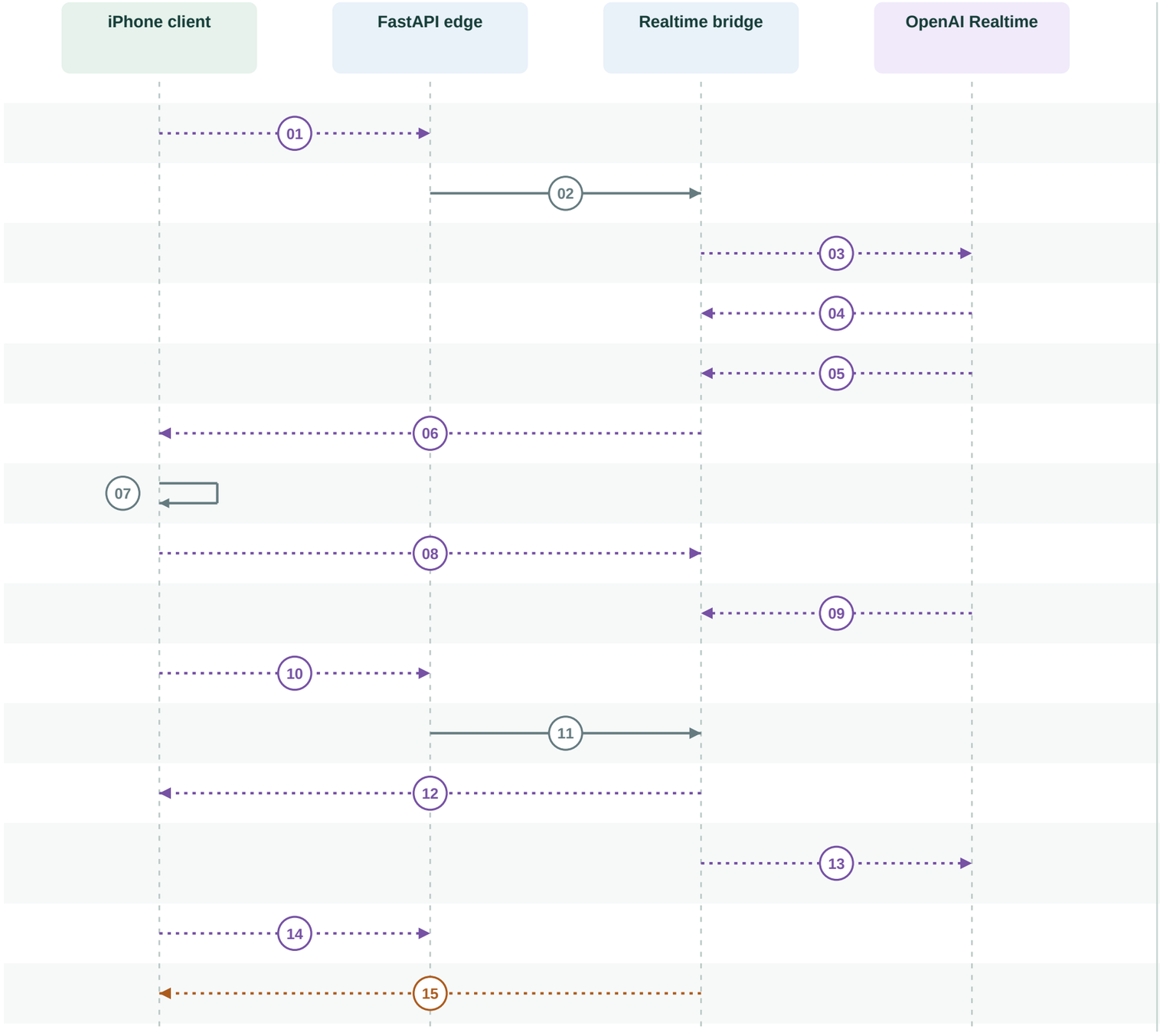
16 **inventory {packet}; later voice_ready**
Initial snapshot is independent of new provider voice readiness.

BOUNDARY Confirmation intent is model-interpreted. Code enforces later intelligible owner turns, gate versions and matching arguments; playback must cover readback and readiness. Preview may run, evidence capture does not.

SOURCE claim_agent/openai_live.py:187 | claim_agent/openai_live.py:426 | claim_agent/onboarding.py:110 | claim_agent/server.py:211 | ios/LibraryClaim/CaptureController.swift:205

F03 Stream audio, acknowledge playback and interrupt

Audio and transcript events stream asynchronously; generation completion does not prove playback.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 audio {pcm_base64}
Signed PCM16 mono at 16 kHz; microphone enabled only after voice_ready.

02 Validate and enqueue audio
Even byte count; at most 32,000 bytes per native audio event.

03 input_audio_buffer.append {audio}
Stateful resampling to 24 kHz; provider VAD controls ordinary turn creation.

04 input_audio_buffer.speech_started / speech_stopped / committed
Independent transcription.completed and response events follow.

05 response.output_audio.delta
PCM16 mono 24 kHz plus response/item/content identifiers.

06 audio via API event socket
pcm_base64, sample_rate=24000, item_id, content_index, audio_end_ms.

07 Schedule playback on AVAudioPlayerNode
Track cumulative played duration with dataPlayedBack callbacks.

08 playback via API
item_id, content_index, played_ms: receipt of audio actually played.

09 Transcript and response.done events
Forward transcript / turn_complete separately; record reported usage.

10 ALT manual: interrupt_voice
Phone stops queued playback and temporarily suppresses late audio.

11 Handle interruption request
Spoken barge-in instead starts from provider input_audio_buffer.speech_started.

12 interrupted via API
Publish interrupted first; invalidate pending work and clear stale playback state.

13 response.cancel; conversation.item.truncate
Manual: response.cancel only if active; truncate only when an unplayed audio tail exists.

14 Optional stop_voice
Close provider bridge while retaining event-only socket where possible.

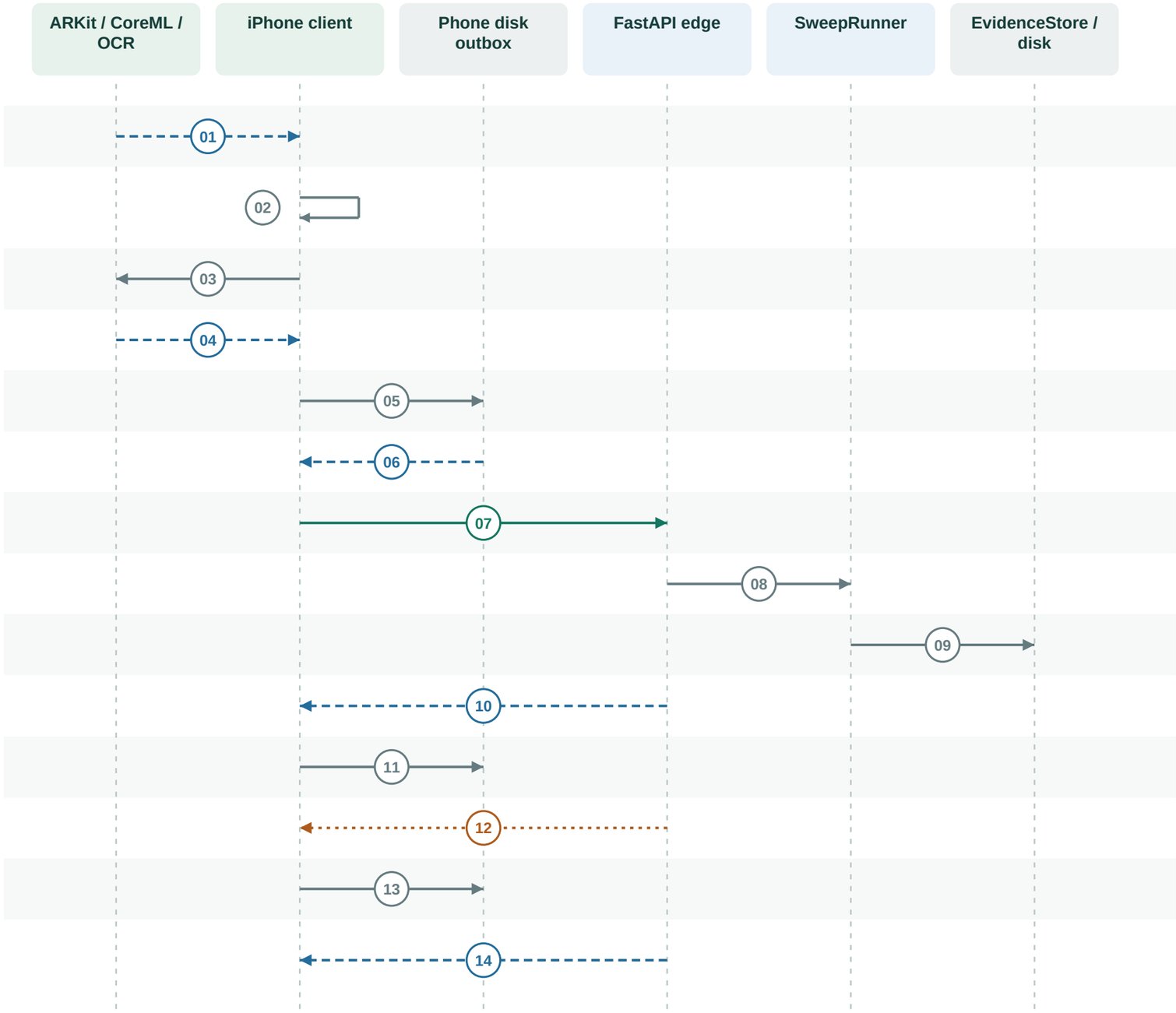
15 ALT unrecovered provider failure: voice_error
Sanitized reconnect/billing guidance; preserve usage and review reason.

BOUNDARY The active protocol has no sequential STT -> text model -> TTS chain. Input transcription assists action gates asynchronously. No routine uploaded JPEG is forwarded to Realtime. Disconnect does not erase saved capture evidence. Native audio and inventory share one client WebSocket; the gateway owns a separate provider WebSocket.

SOURCE ios/LibraryClaim/LiveAudio.swift:53 | ios/LibraryClaim/LiveAudio.swift:135 | claim_agent/server.py:448 | claim_agent/openai_live.py:286 | claim_agent/openai_live.py:495

F04 Select, retain and upload camera evidence

An HTTP receipt acknowledges immutable admission or suppression. Recognition completes later (F05).



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 Local detector proposals + quality + fingerprint
Core ML full frame/detail crops; display provisional dotted outlines.

02 Apply selection and concurrency gates
Tracking, blur/exposure, novelty, pending count, time interval, depth improvement.

03 Encode selected frame and run Vision OCR
Raw-axis JPEG + calibration/pose + OCR; LiDAR depth/confidence if supported.

04 Encoded FrameUpload payload
local_analysis=cloud_requested/unavailable; no Apple language model call.

05 Atomically save pending/{frame_id}.json
Immutable bytes before network; 1 GB outbox budget.

06 Earliest pending frame payload
One uploader sends saved frames sequentially; retries reuse exact bytes.

07 POST /api/sweeps/{id}/frames
metadata, jpeg_base64, depth_base64, confidence_base64; bearer auth.

08 Validate payload and compute receipt signature
Dimensions, image/depth lengths, same AR session and capture state.

09 Persist evidence, receipt and queued state
New frames enter bounded queue; JPEG content is hash-addressed.

10 200 {accepted:true, processing:"queued", ...}
duplicate:false and queue_depth for new frames; not a recognition result.

11 Move pending file to uploaded/
Preserve bytes; accepted frame unlocks live geometry upload.

12 ALT full queue: 429 + Retry-After: 3
detail.processing=retry_backpressure, accepted:false, retry_after_s:3.

13 ALT retain pending; delay and retry same bytes
Other upload errors also retain pending evidence; no second frame ID.

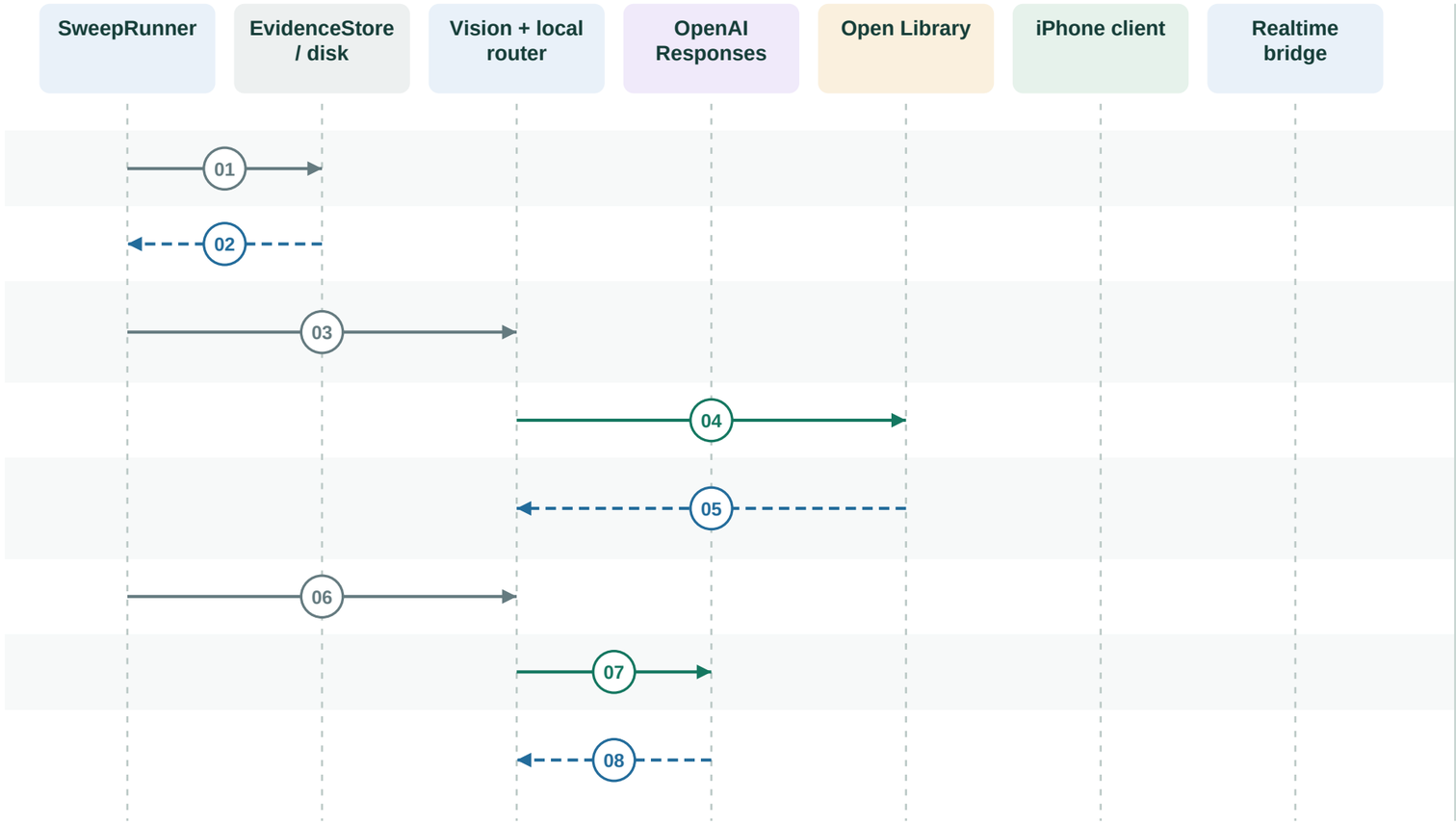
14 ALT duplicate: 200 with duplicate:true
processing=duplicate_suppressed or queued_geometry; original receipt may replay.

BOUNDARY The phone retains acknowledged bytes. Same-ID retries cannot change metadata/calibration. Similar-image suppression stores metadata, not a second JPEG; exact RGB plus new depth may queue geometry-only refinement.

SOURCE ios/LibraryClaim/LocalRecognitionModels.swift:142 | ios/LibraryClaim/FrameEncoder.swift:48 | ios/LibraryClaim/EvidenceOutbox.swift:23 | claim_agent/pipeline.py:154 | claim_agent/server.py:361

F05 Recognize a frame and publish inventory - part 1/2

Frame workers run concurrently; publication order can differ from capture order.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 Dequeue frame; read hash-verified JPEG
Duplicate-geometry job instead reuses source polygons with its own depth.

02 Saved image + calibrated evidence
Receipt integrity establishes bytes, not recognition correctness.

03 Resolve local OCR / proposal evidence
Apply independent text and detector support; optional catalogue lookup precedes cloud reservation.

04 OPTIONAL GET openlibrary.org/search.json
Compatible book clues only: title, author, limit=3, selected metadata fields.

05 OPTIONAL candidate bibliographic metadata
Corroboration only; catalogue ISBN/edition does not become observed evidence.

06 Quality / novelty / budget / cooldown decision
Reserve and persist any cloud attempt before a billable provider request.

07 POST /v1/responses (standard OpenAI base)
gpt-6-astra; upright selected JPEG + hints; store:false; strict ProposedScan.

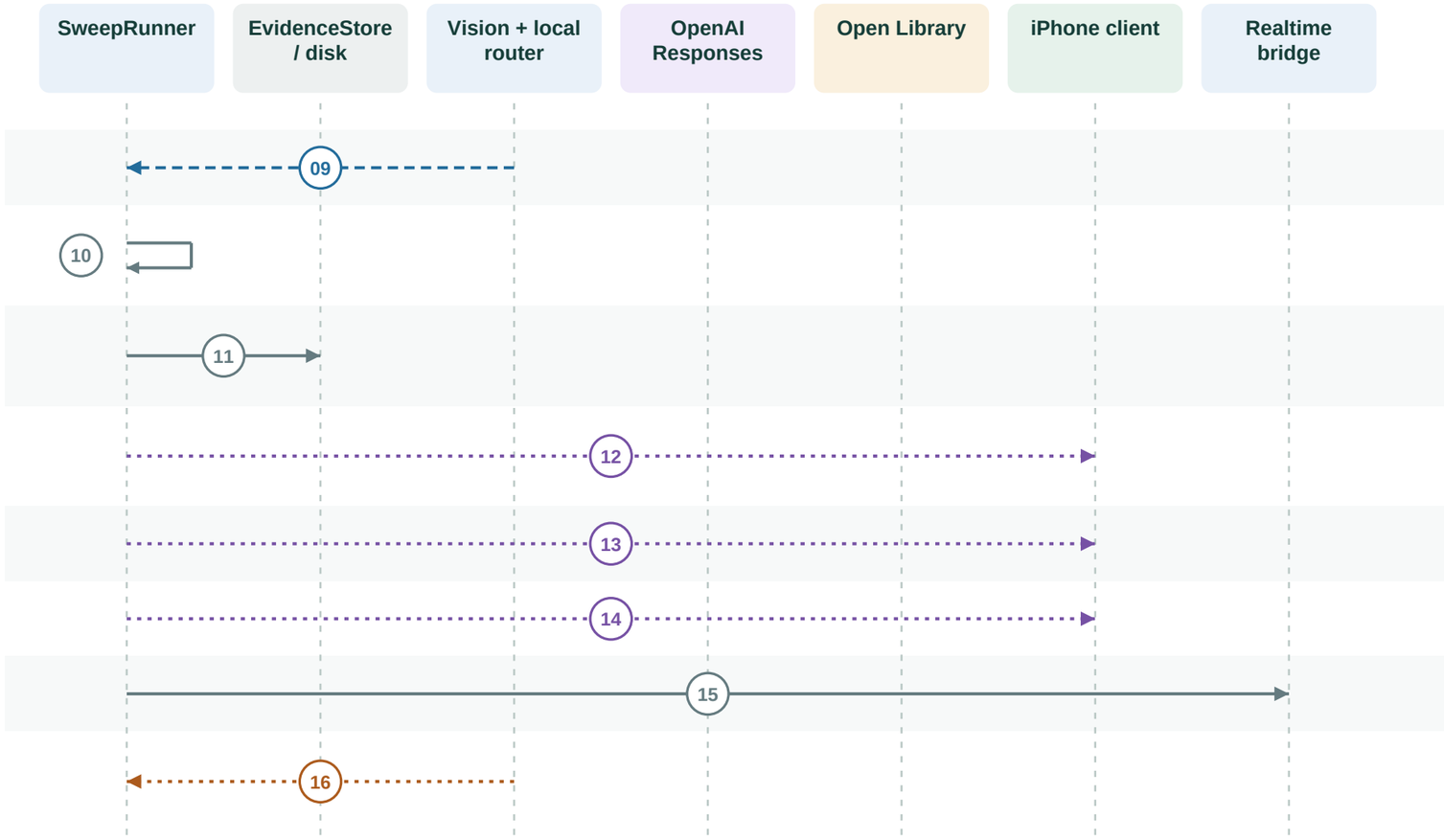
08 Completed parsed output + usage
Detections, polygons, visible text, quality issues, guidance; no price/scale authority.

BOUNDARY Current phone produces no local semantic book clues; Open Library is a compatibility branch, not the normal current path. Quality/budget/cooldown can defer cloud. Model labels do not set dimensions or prices.

SOURCE claim_agent/pipeline.py:237 | claim_agent/local_evidence.py:67 | claim_agent/local_evidence.py:161 | claim_agent/openai_provider.py:24 | claim_agent/pipeline.py:286

F05 Recognize a frame and publish inventory - part 2/2

Frame workers run concurrently; publication order can differ from capture order.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

09 Validated scan + routing + quality
Restore raw-image coordinates; reject malformed individual polygons.

10 Under lock: geometry, association and apply_scan
Measure sensor-supported faces; match physical records; gate book identity on OCR.

11 Save observation/overlays; revise state
Increment revision, persist immutable refs and atomic state; queue eligible prices.

12 inventory {packet} via API WebSocket
Full saved snapshot, including usage/cost estimate and unresolved review entries.

13 frame_overlays then recognition_status
Overlay geometry; status ready/pending, routing, cost and provisional objects.

14 Optional guidance via API WebSocket
User-facing capture guidance, separate from structured voice context.

15 offer_context: saved recognition facts
Grounded text context for Realtime (F03); no routine JPEG copied to voice.

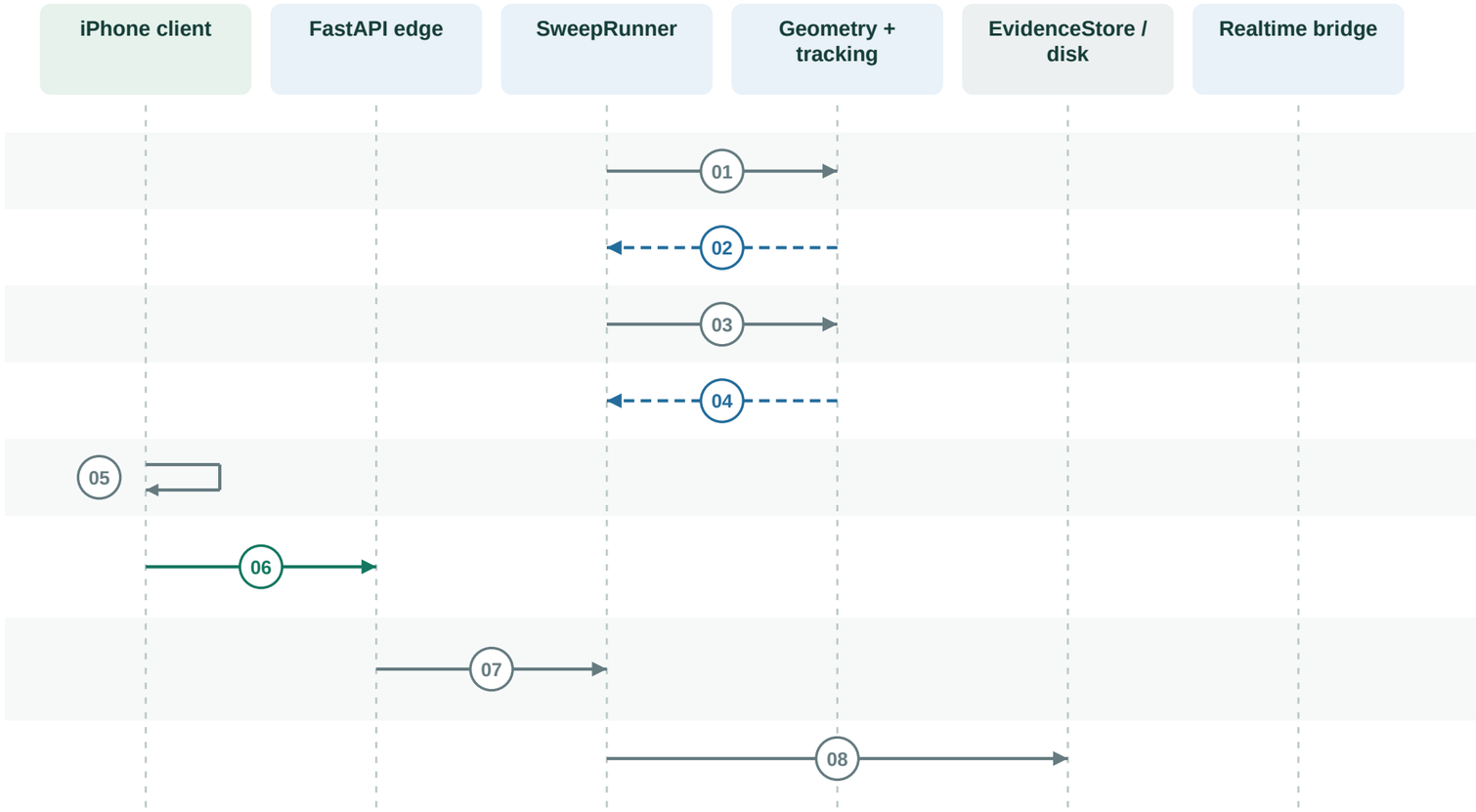
16 ALT deferred / provider failure / worker failure
Retain supported evidence; unknown usage is null; emit pending or unavailable status.

BOUNDARY Current phone produces no local semantic book clues; Open Library is a compatibility branch, not the normal current path. Quality/budget/cooldown can defer cloud. Model labels do not set dimensions or prices.

SOURCE claim_agent/pipeline.py:237 | claim_agent/local_evidence.py:67 | claim_agent/local_evidence.py:161 | claim_agent/openai_provider.py:24 | claim_agent/pipeline.py:286

F06 Measure objects and update RoomPlan geometry - part 1/2

Object face measurements and room/furniture geometry use calibrated sensor evidence, not guessed model scale.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 For each scan polygon: measure detection
Use original calibration and confident aligned depth; fit local plane/rays.

02 Dimensions / world support or unknown + reasons
Book spine edges or visible object face; no invented full 3-D box.

03 Associate physical track and furniture anchor
Same session, one-to-one assignment, compatible category/depth/geometry.

04 Existing physical ID or new/reviewable record
Text equality alone never merges physical objects; ambiguity stays explicit.

05 RoomPlan live update on shared AR session
Throttle >=3 s; retain latest snapshot; only one geometry request in flight.

06 POST /api/sweeps/{id}/geometry
After accepted frame: session ID, walls/openings/objects; live floors:[], capture_complete:false.

07 Validate session and geometry write state
Camera mode, absent first frame or mismatched session reject; same bytes are idempotent.

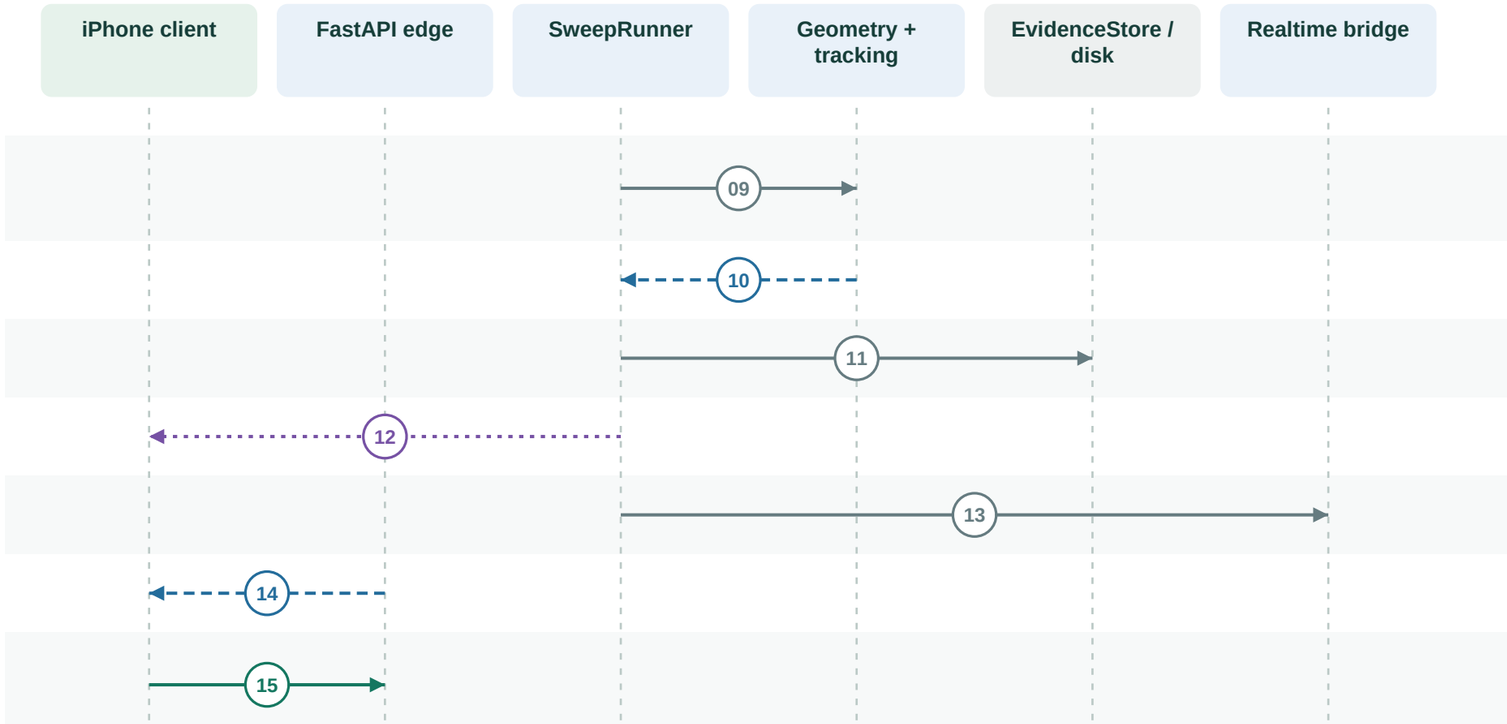
08 Save immutable geometry/room_<id>.json
Preserve the input geometry before reconciling measurements.

BOUNDARY Live RoomPlan intentionally omits floor polygons. Final RoomBuilder geometry is needed for floor area. Camera-only frames keep unsupported measurements unknown. Accurate sensor-derived numbers still need tape/ruler validation.

SOURCE claim_agent/pipeline.py:369 | claim_agent/geometry.py:30 | ios/LibraryClaim/MeasurementModels.swift:4 | ios/LibraryClaim/RoomGeometry.swift:5 | claim_agent/pipeline.py:711

F06 Measure objects and update RoomPlan geometry - part 2/2

Object face measurements and room/furniture geometry use calibrated sensor evidence, not guessed model scale.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

09 measure_room + reconcile furniture
Floor/wall polygons, openings, wall height, supported RoomPlan boxes, one-to-one matches.

10 Room facts / furniture dimensions / review issues
Curved or incomplete surfaces and ambiguous links preserve unknowns.

11 Update state; refresh affected price queries
Clear obsolete dimensions on revision; stale quotes cannot apply.

12 inventory {packet} via API WebSocket
Event update can arrive before the geometry HTTP response.

13 offer_context: measurement facts
Optional grounded voice cue; this is an internal bridge call, not a phone event.

14 200 {saved:true}
422 unsupported/session-invalid; 409 changed geometry after capture ended.

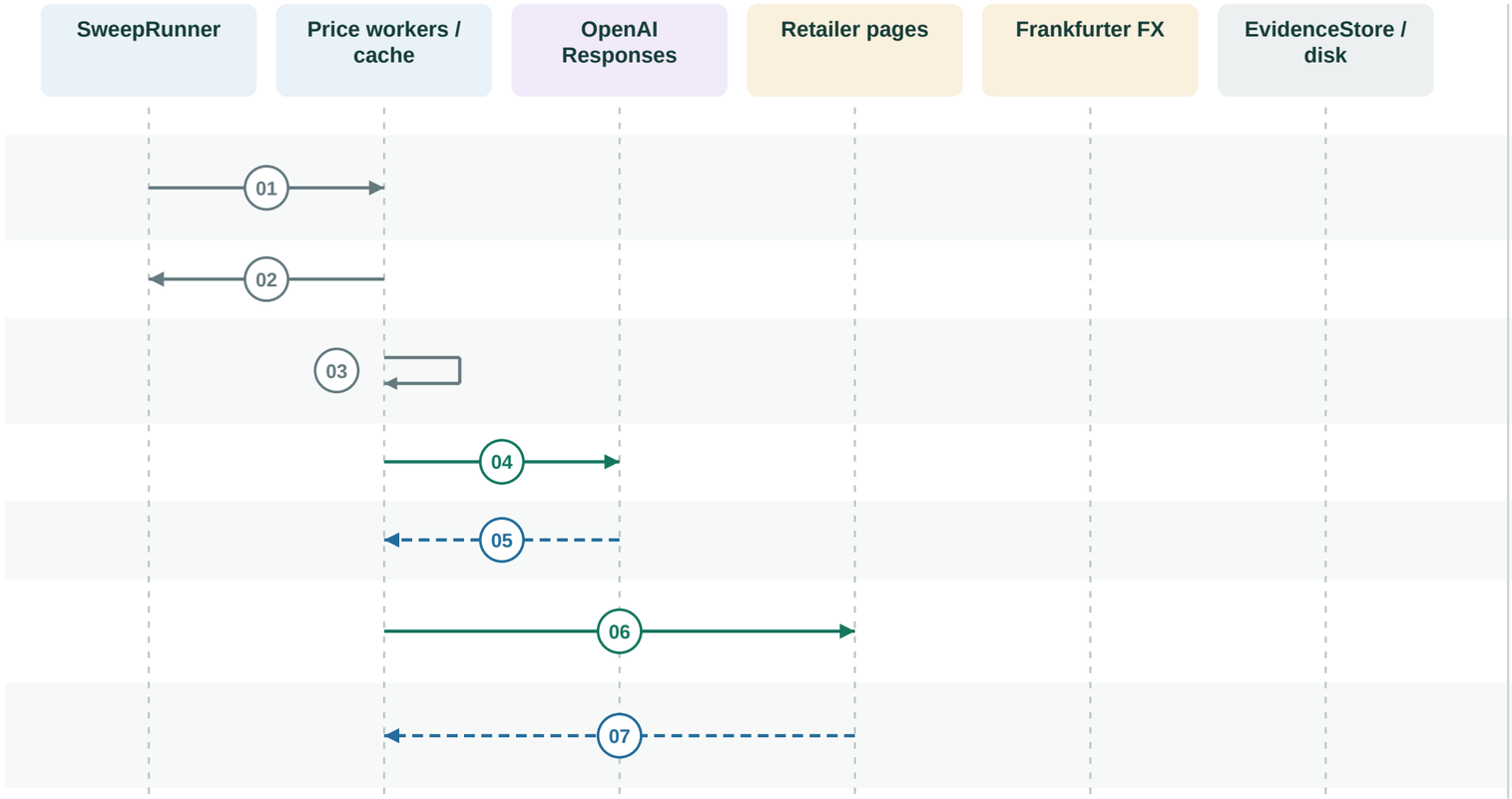
15 At end, if available: POST final geometry
LiDAR RoomBuilder output before /finish; failures become capture issues and geometry may be skipped.

BOUNDARY Live RoomPlan intentionally omits floor polygons. Final RoomBuilder geometry is needed for floor area. Camera-only frames keep unsupported measurements unknown. Accurate sensor-derived numbers still need tape/ruler validation.

SOURCE claim_agent/pipeline.py:369 | claim_agent/geometry.py:30 | ios/LibraryClaim/MeasurementModels.swift:4 | ios/LibraryClaim/RoomGeometry.swift:5 | claim_agent/pipeline.py:711

F07 Search, verify and apply a price - part 1/2

Pricing is a separate in-process queue. Search returns URLs; fetched source offers establish price evidence.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 Enqueue eligible record + revision/fingerprint
Books: new and used separately; items: new only. No repeat of unchanged attempted job.

02 Set queued -> searching; save/publish inventory
Check eligibility, current revision, paid-pricing setting and provider latch.

03 Check quote/search cache and shared in-flight task
Quote cache skips network; search cache skips model search only. Per-runner caches; raw current query validated.

04 POST /v1/responses with web_search
gpt-5.4-mini, country/currency/condition query, required search, one tool call.

05 Completed search citations/sources + usage
Up to eight candidate URLs; generated price prose is not accepted.

06 GET discovered public product URLs
Fetch candidates concurrently; revalidate redirects/addresses; timeout/size limits.

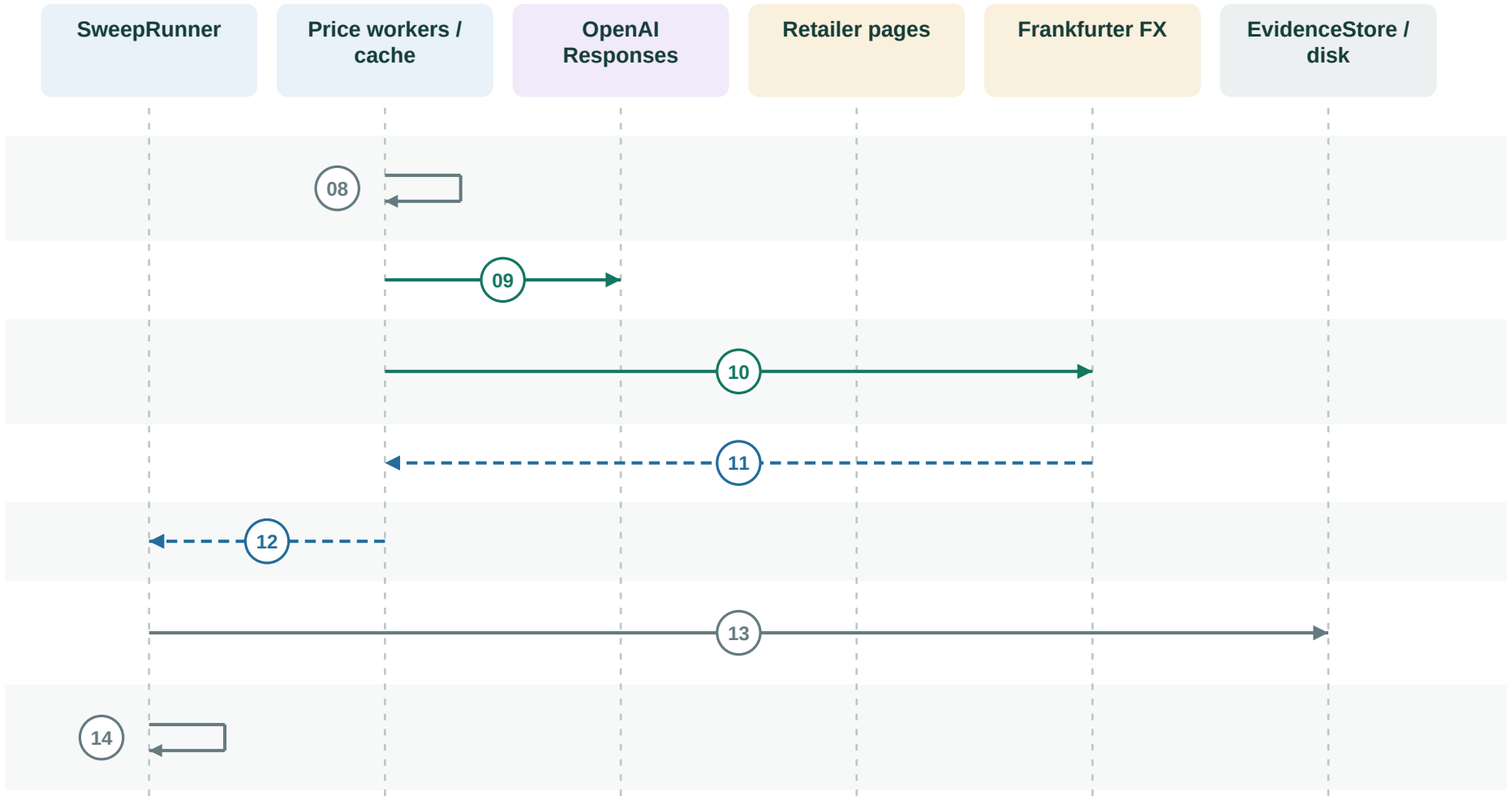
07 HTML/text containing JSON-LD Product/Book Offer
No fixed retailer pricing API; blocked/unstructured sources may give no usable offer.

BOUNDARY Quote-cache hit skips network calls; search-cache hit skips only model search and still fetches/validates retailers. Caches/in-flight sharing are per runner. A quote has a 45 s deadline after semaphore entry. Unknown offers stay outside totals. API USD usage costs are separate from local-currency claim values.

SOURCE claim_agent/pipeline.py:551 | claim_agent/pipeline.py:620 | claim_agent/pricing.py:350 | claim_agent/openai_provider.py:71 | claim_agent/providers.py:18 | claim_agent/pricing.py:466

F07 Search, verify and apply a price - part 2/2

Pricing is a separate in-process queue. Search returns URLs; fetched source offers establish price evidence.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

08 Validate and rank individual offers
Identity, condition, stock, market/currency, material/dimensions and retrieval date.

09 OPTIONAL foreign-market fallback search
If local sources fail, search UK/GBP or India/INR; repeat retrieval/validation.

10 OPTIONAL GET /v2/rate/{base}/{quote}
Foreign conversion and non-INR appraisal-threshold check; pair cache may satisfy.

11 JSON {base, quote, date, rate}
Validate dated positive rate and pair; preserve original amount/currency.

12 QuoteResult with Price and source metadata
Or amount:null + explicit unavailable/timeout/access reason.

13 Revalidate revision; persist offer/search/FX refs
Reject stale results; update pricing status; no automatic retry after uncertain spend.

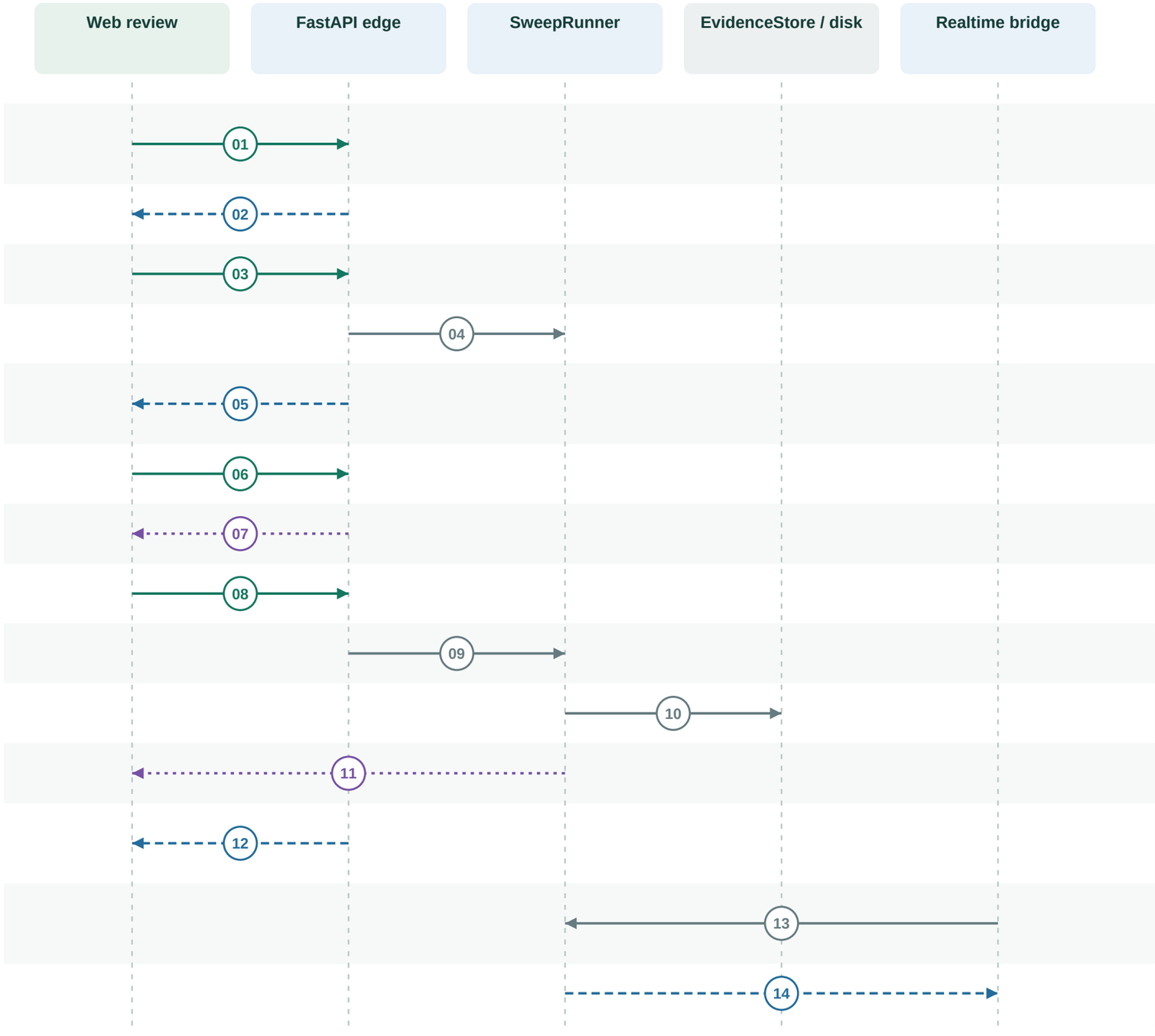
14 apply_quote -> save_publish -> optional bundle refresh
Later inventory event and valuation_updated voice context announce current state.

BOUNDARY Quote-cache hit skips network calls; search-cache hit skips only model search and still fetches/validates retailers. Caches/in-flight sharing are per runner. A quote has a 45 s deadline after semaphore entry. Unknown offers stay outside totals. API USD usage costs are separate from local-currency claim values.

SOURCE claim_agent/pipeline.py:551 | claim_agent/pipeline.py:620 | claim_agent/pricing.py:350 | claim_agent/openai_provider.py:71 | claim_agent/providers.py:18 | claim_agent/pricing.py:466

F08 Read inventory and apply corrections

Browser review pairs with a cookie; native HTTP uses a bearer token. Both can subscribe to the same event route.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 POST /api/pair {token}
Optional browser bootstrap; token supplied from review link fragment, then cleared.

02 200 {paired:true} + claim_pair cookie
HttpOnly, SameSite=Strict; Secure only over HTTPS. Bad token: 401.

03 GET /api/sweeps; GET /api/sweeps/{id}
List latest 20 saved entries; select one for a current snapshot.

04 runner_for(id): lazy load if needed
After restart mark interrupted capture partial; resume eligible pricing only.

05 200 list / 200 ClaimPacket
Packet includes sweep, room, books/items, totals, review_queue, revision, metrics.

06 WS /api/sweeps/{id}/live
Browser can stay event-only; it does not send start_voice.

07 Initial inventory, then updated packets/events
Initial snapshot and later inventory use {type:"inventory",packet}.

08 POST /api/sweeps/{id}/corrections
JSON: action, ref_id, value. Use an exact saved inventory/shelf ID.

09 correct(): resolve alias, validate and revise
Invalidate affected price jobs/values; append owner claim/exclusion audit.

10 Persist revised state and refresh finished bundle
Print confirmation may enable pricing; unsupported claims require appraisal.

11 inventory {packet} via API
State publication occurs before the correction HTTP response.

12 200 updated ClaimPacket
Invalid reference/action: 422; identical correction is not automatically deduplicated.

13 ALTERNATIVE voice: correct_inventory tool
Validated provider function event invokes the same method, not another HTTP service.

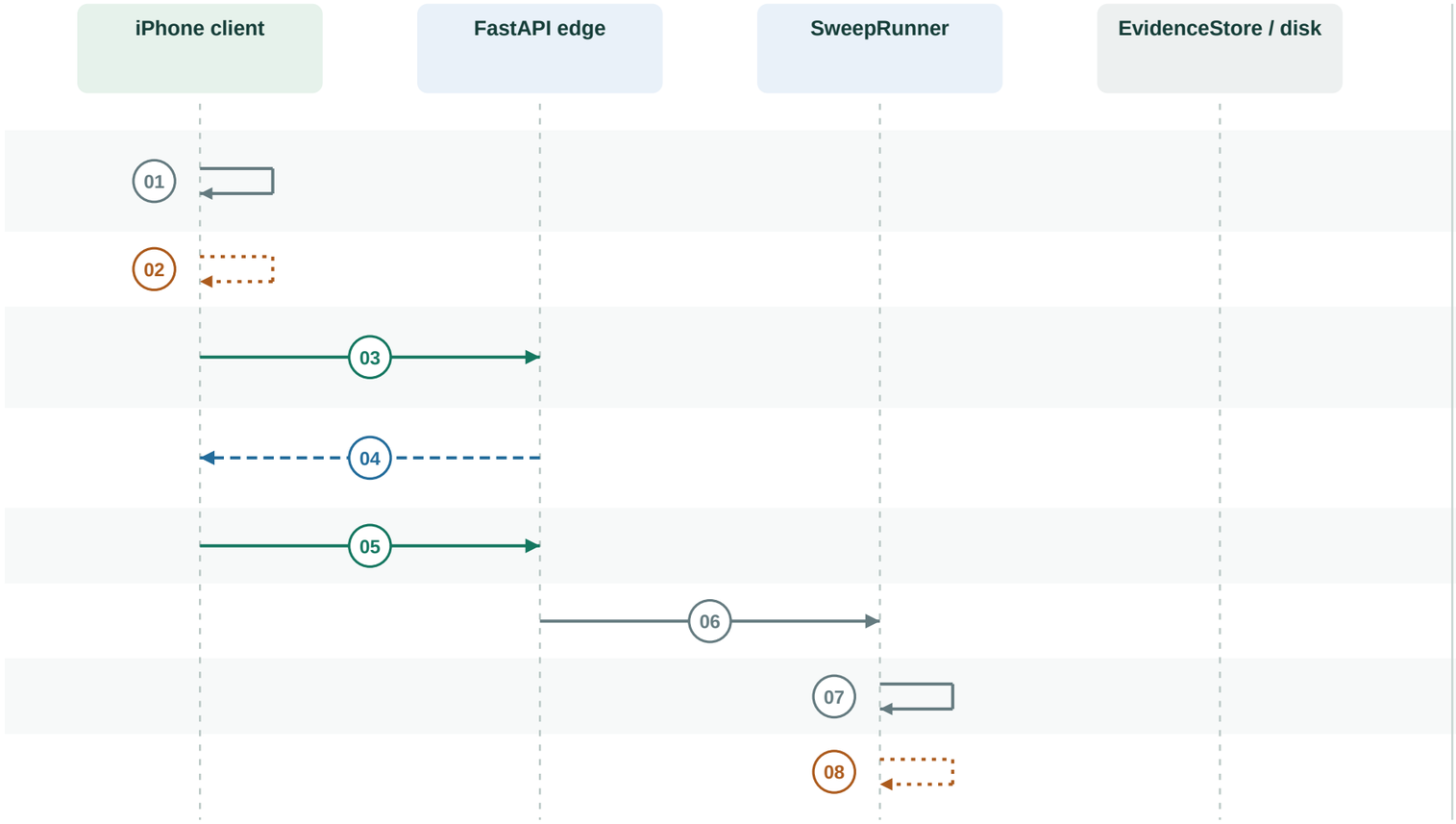
14 Internal snapshot/result -> function_call_output
Tool result is sent back to Realtime by call_id; see F02/F03.

BOUNDARY There is no SSE endpoint or durable event cursor. Listener overflow drops oldest events; GET or a new WebSocket snapshot recovers current state. Lazy-loading a saved sweep may resume eligible pricing, but not frame jobs.

SOURCE claim_agent/static/app.js:28 | claim_agent/server.py:157 | claim_agent/server.py:193 | claim_agent/server.py:378 | claim_agent/pipeline.py:732 | claim_agent/live.py:65

F09 Finish, generate and download claim evidence - part 1/2

The native client flushes saved evidence and final geometry before a held /finish request.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 Stop sampling; LiDAR RoomBuilder if supported
Await encoder; save final geometry if available; drain pending uploads for up to 60 s.

02 ALT pending evidence remains: Awaiting uploads
Expose Retry finish; do not call server finish while pending files remain.

03 If available: POST /geometry after live drain
Send final LiDAR geometry when present; camera or failed export skips it with capture issues.

04 200 {saved:true}
Require success when geometry exists; otherwise finalize with recorded geometry-failure issues.

05 POST /api/sweeps/{id}/finish
duration_s and capture_issues; native request timeout 300 s.

06 finish_lock; transition to processing
Already ready/partial returns snapshot. Otherwise save/publish processing state.

07 Await frame queue, then price queue
Workers run concurrently; bounded overall finish_timeout, source default 240 s.

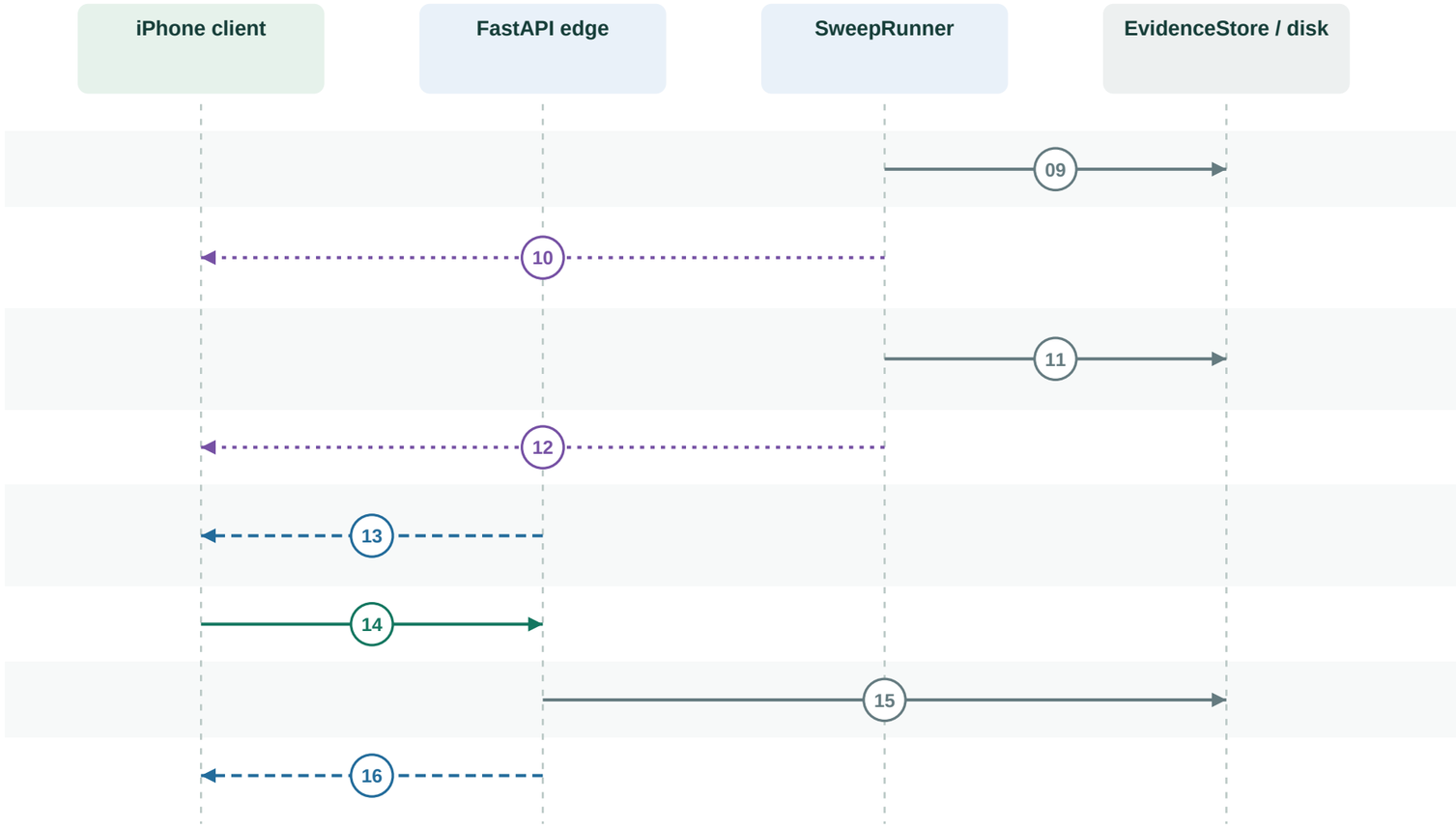
08 ALT deadline: cancel remaining workers
Mark unfinished pricing unknown/reviewable; continue to partial packet.

BOUNDARY Finish returns 200 only after drain/deadline handling; it is not a 202 job receipt. Partial can still be a successful HTTP outcome. Exports may also be requested before finish. Comparison is not automatically inside packet.zip.

SOURCE ios/LibraryClaim/CaptureController.swift:487 | ios/LibraryClaim/CaptureController.swift:519 | claim_agent/pipeline.py:776 | claim_agent/packet.py:150 | claim_agent/server.py:397

F09 Finish, generate and download claim evidence - part 2/2

The native client flushes saved evidence and final geometry before a held /finish request.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

09 Compute Decimal totals; save final state
Preserve ready/partial state, review queue and immutable evidence refs.

10 inventory {packet} via API
Final snapshot is published before bundle generation; it does not prove the ZIP exists.

11 Build bundle; update timing; save/rewrite bundle
Generate JSON, HTML and ZIP; then offer capture-ended context to internal voice bridge.

12 packet_ready {packet} via API
Emitted after bundle generation; final HTTP response follows.

13 200 final/partial ClaimPacket
Held HTTP response completes; this does not prove all acceptance bars passed.

14 GET /api/sweeps/{id}/packet.zip
Optional alternatives: claim_packet.json, report.html, evidence path.

15 Rebuild bundle under lock; verify evidence hashes
ZIP includes packet/report/manifest and registered evidence, not raw state.json.

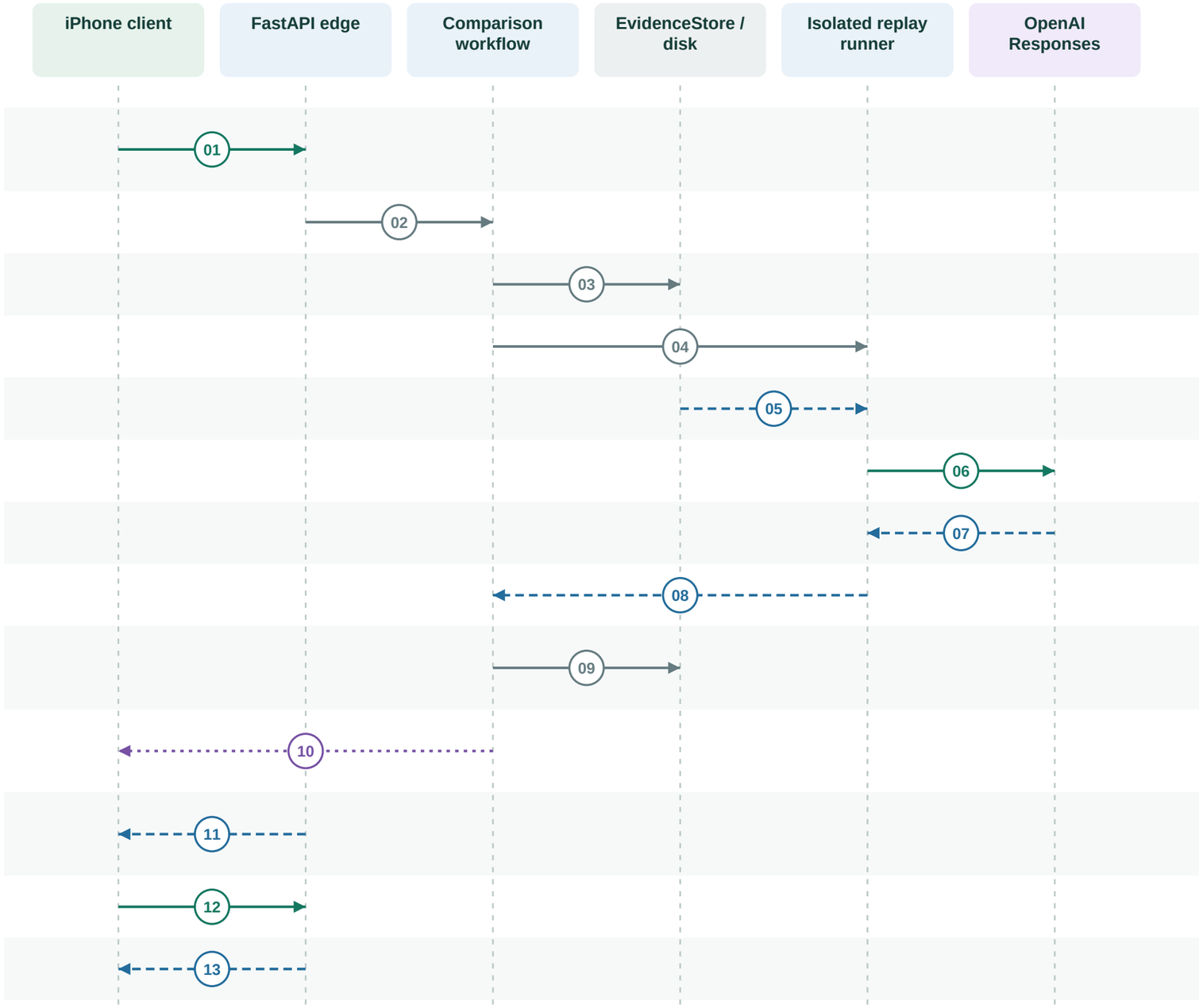
16 200 application/zip; filename claim_{id}.zip
Native saves local download; optional comparison starts afterwards (F10).

BOUNDARY Finish returns 200 only after drain/deadline handling; it is not a 202 job receipt. Partial can still be a successful HTTP outcome. Exports may also be requested before finish. Comparison is not automatically inside packet.zip.

SOURCE ios/LibraryClaim/CaptureController.swift:487 | ios/LibraryClaim/CaptureController.swift:519 | claim_agent/pipeline.py:776 | claim_agent/packet.py:150 | claim_agent/server.py:397

F10 Compare camera-only with LiDAR

Optional same-frame depth ablation: one held POST, persisted status reads, and a separate replay runner.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 POST /api/sweeps/{id}/camera-comparison
JSON {}; native waits up to 300 s. Must follow eligible ready/partial LiDAR sweep.

02 Acquire filesystem lock; inspect previous report
Existing completed/failed report returns without another replay.

03 Freeze LiDAR packet; persist running status
Validate model/provider continuity, evidence and total active capacity.

04 Create fresh camera runner / provider / state
Same locale/model and original RGB+calibration/pose; zero depth dimensions.

05 Original hash-verified JPEGs and calibration
Do not inherit source tracks, physical IDs, corrections, geometry or prices.

06 Gated vision/search requests (F05/F07)
Independent call budget and costs; process each frame, then finish replay.

07 Independent observations / sources / usage
Unknown/error outcomes remain distinct from zero detected objects.

08 Camera packet and processing evidence
Counts, measurement coverage, review reasons, timing and usage.

09 Persist camera packet and comparison report
status complete or failed; analysis_status separately completed/partial/failed/interrupted.

10 comparison_ready {report} via API
Completed/handled failure emits event. Cancellation may persist failure without event; recover through GET.

11 200 saved report after processing
No 202 response. 409 already_running/source_unavailable/capacity before replay.

12 OPTIONAL GET /api/sweeps/{id}/camera-comparison
Read persisted status; 404 if absent. No replay starts from this GET.

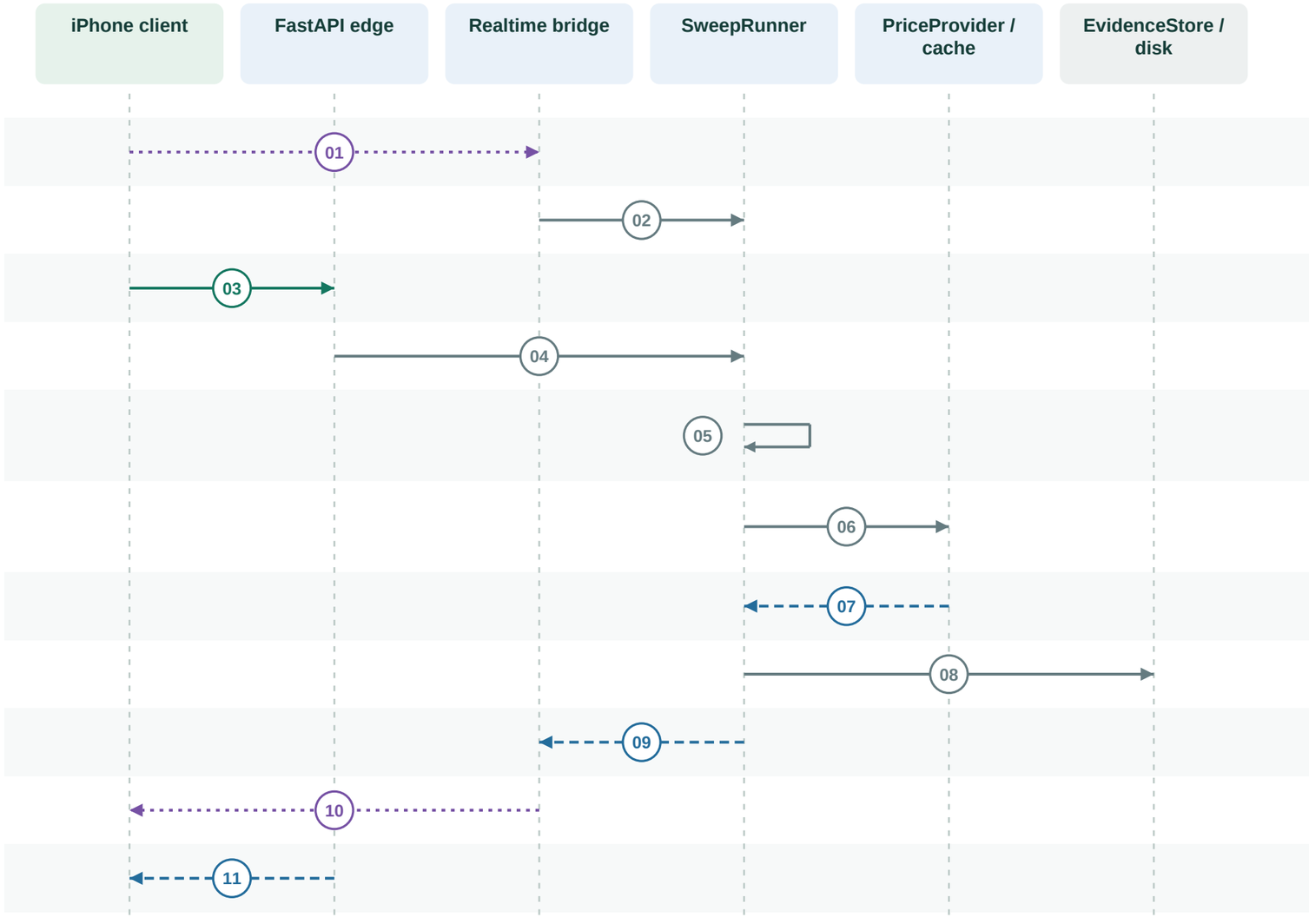
13 200 persisted running/complete/failed report
Already-running native recovery does one GET; no automatic polling loop.

BOUNDARY This reuses JPEGs and AR poses, not source depth/geometry/identity/prices. Poses may have benefited from LiDAR. A complete workflow can contain partial/failed analysis; it is not an accuracy or sensor-speed benchmark.

SOURCE claim_agent/comparison.py:107 | claim_agent/comparison.py:115 | claim_agent/comparison.py:152 | ios/LibraryClaim/BackendClient.swift:119

F11 Reprice the same books in a second market

Two entry points share one method: voice emits a result event; direct HTTP returns the result without that event.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 PATH A: spoken request through Realtime (F03)
Validated compare_market function arguments: country, currency.

02 compare_market -> second_locale(locale)
Validate supported voice country/currency pair; no separate HTTP round trip.

03 PATH B: POST /api/sweeps/{id}/second-locale
JSON: country, currency; request stays open until results are ready.

04 second_locale(locale)
This is an alternative entry to step 2, not a second call after voice.

05 Copy first ten identified, nonexcluded books
Freeze same physical IDs and frame refs; report required=10 and actual coverage.

06 For each selected book: quote new, then used
Direct sequential quote calls with F07 validation; bypasses pricing-job queue and its resume/dedup rules.

07 Prices with sources or explicit unknowns
Do not overwrite primary-market prices or generate missing identities.

08 Save source evidence and second-locale observation
Persist state; no inventory revision/event or bundle regeneration here.

09 PATH A result: same_physical_ids:true + books
Bridge receives result after awaited pricing.

10 PATH A only: second_locale {result}
Also return function output to Realtime so spoken response can use result.

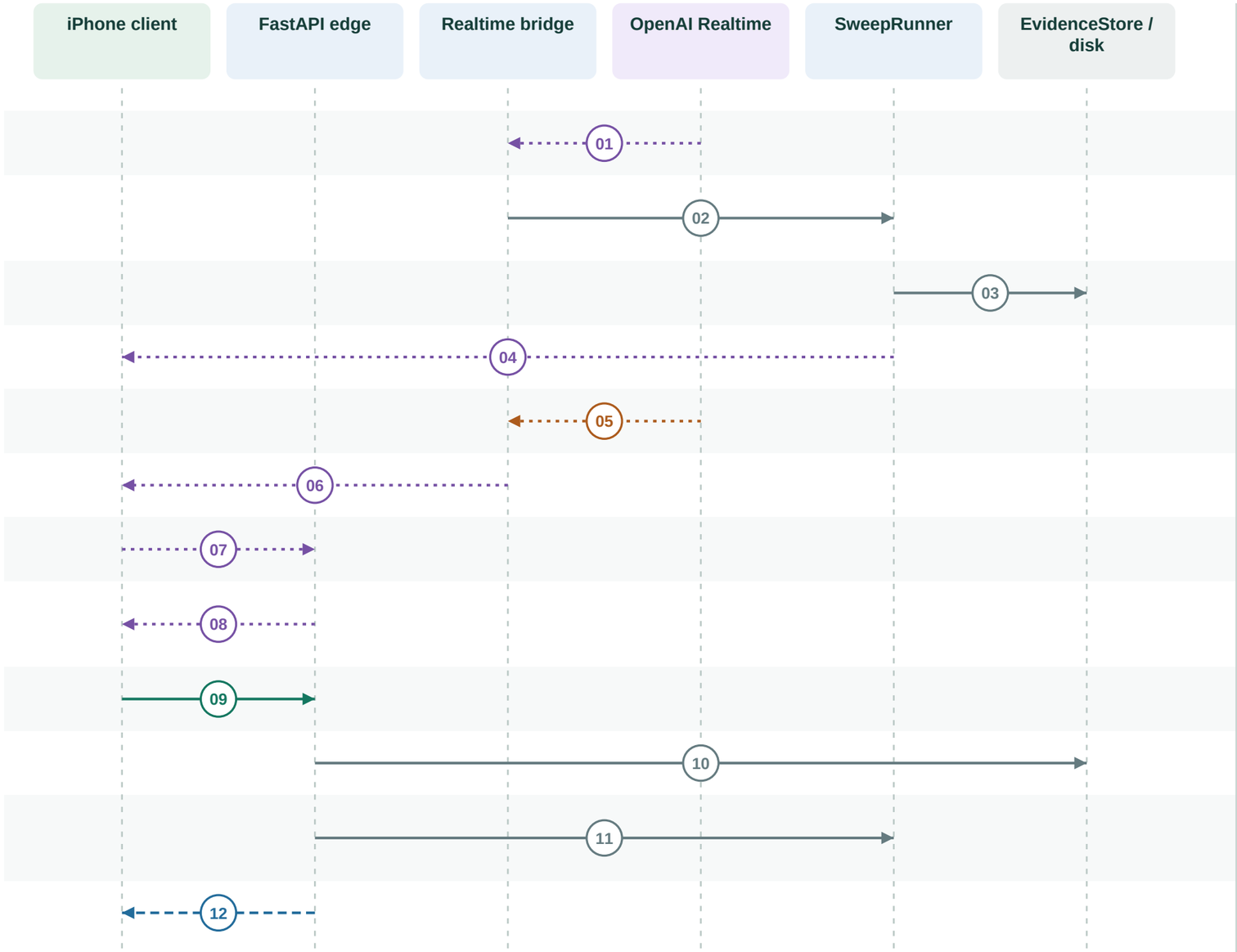
11 PATH B only: 200 second-locale result
No second_locale WebSocket event is emitted by this HTTP route.

BOUNDARY Copies the first ten eligible identified books, or fewer with coverage. Original IDs/frame refs stay the same; primary market prices are unchanged. Direct HTTP validates Locale syntax; voice also validates supported market pairing.

SOURCE claim_agent/server.py:394 | claim_agent/live.py:73 | claim_agent/pipeline.py:858 | ios/LibraryClaim/CaptureController.swift:698

F12 Reconnect, recover and account for usage

Recovery is bounded and explicit. Files are durable; provider sessions and most processing queues are not.



MESSAGE / PAYLOAD / RESULT

Read downward; numbers match the arrows.

01 **response.done with response.usage; transcription usage**
Record actual provider usage once per model/response ID.

02 **persist_usage -> save_publish**
Sweep path persists metrics; onboarding usage stays in memory until copied at handoff. Unknown usage:null.

03 **Atomic state + cost ledger snapshot**
Compute known estimated subtotal and uncosted count; unknown is not zero.

04 **inventory {packet} via API**
Metrics are nested inside packet; no dedicated usage event.

05 **ALT provider access/credit or fatal voice error**
Selected stale control errors recover; other failures terminate voice bridge.

06 **voice_error; review reason saved**
Request explicit reconnect/funding action; retain saved camera evidence.

07 **User reconnect: close old WS, open new WS**
Repeat health/readiness negotiation in F01; no background paid reconnect loop.

08 **Current snapshot and newly negotiated voice**
Unexpired in-memory setup can survive reconnect, not server restart. Prior provider audio buffers do not.

09 **After server restart: GET saved sweep**
runner_for loads state; saved capturing/processing changes to partial.

10 **Load persisted state and evidence**
No durable frame-task broker; missing capture continuity is reviewable.

11 **resume_pricing with identity/spend guards**
Queued eligible jobs may resume; searching becomes interrupted without replay.

12 **200 current ClaimPacket with review/unknowns**
Later price publications use the same event route; no claimed full recovery.

BOUNDARY No blanket automatic paid retry. Persisted searching price jobs become interrupted/unknown-cost; eligible queued prices may resume. Server restart does not replay saved frame jobs. Costs can remain partial even when a packet exists.

SOURCE claim_agent/server.py:157 | claim_agent/pipeline.py:61 | claim_agent/openai_live.py:238 | claim_agent/openai_live.py:694 | claim_agent/costing.py:57 | ios/LibraryClaim/CaptureController.swift:742

Endpoint and contract index

Active routes and optional workflows; legacy voice HTTP routes remain implemented but are not used by the active native Realtime flow.

Method	Route	Result / transport	Purpose / flow
GET	/api/health	200 configuration JSON	F01; unauthenticated; no paid readiness probe
POST	/api/pair	200 {paired:true}; cookie	F08 browser-only pairing; 401 for bad token
POST	/api/onboarding	200 {id,expires_in_s}; 409 capacity	F01; JSON device/capture_supported/camera_supported
WS	/api/onboarding/{id}/live	Accepted + async setup/audio events	F01-F03; 4401/4404 are pre-accept rejection branches
POST	/api/sweeps	200 ClaimPacket; 409 invalid handoff	F02; native {device,onboarding_id}; retry reuses saved handoff
GET	/api/sweeps	200 summary array, latest 20	F08; reads state files, does not instantiate runners
GET	/api/sweeps/{id}	200 current ClaimPacket	F08/F12; lazy load may resume eligible pricing
WS	/api/sweeps/{id}/live	Initial inventory; async events	F03/F05/F08; no SSE; voice optional on this socket
POST	/api/sweeps/{id}/frames	200 receipt / 429 backpressure	F04; metadata + JPEG/depth/confidence base64; result later via WS
POST	/api/sweeps/{id}/geometry	200 {saved:true}	F06; same-session geometry; invalid/session errors 422
POST	/api/sweeps/{id}/corrections	200 updated ClaimPacket	F08; {action,ref_id,value}; inventory published before response
POST	/api/sweeps/{id}/finish	200 ready/partial ClaimPacket	F09; held request, not 202; duration_s/capture_issues
POST	/api/sweeps/{id}/second-locale	200 comparison result	F11; held request; does not itself emit second_locale event
GET	/api/sweeps/{id}/camera-comparison	200 persisted report; 404 absent	F10; reads running/complete/failed; does not start replay

Common errors: 401 pairing, 403 Origin, 413 body size, 422 validation; operation-specific alternatives appear in each flow.

Endpoint and contract index

Active routes and optional workflows; legacy voice HTTP routes remain implemented but are not used by the active native Realtime flow.

Method	Route	Result / transport	Purpose / flow
POST	/api/sweeps/{id}/camera-comparison	200 report; 409 conflict/capacity	F10; held request; reuses saved result without extra replay
GET	/api/sweeps/{id}/packet.zip	200 application/zip attachment	F09; rebuild current bundle; not gated on finished state
GET	/api/sweeps/{id}/claim_packet.json	200 current ClaimPacket JSON	F09; computed snapshot rather than reading stored file
GET	/api/sweeps/{id}/report.html	200 text/html	F09; regenerate bundle then serve readable report
GET	/api/sweeps/{id}/{kind}/{name}	200 JPEG or octet-stream	F09; allowlisted path and saved SHA-256 validation
GET/POST	/api/{onboarding|sweeps}/{id}/local-voice	Legacy context / tool contract	Inactive native path; do not confuse with OpenAI Realtime
POST	/api/{onboarding|sweeps}/{id}/local-voice/assistant	Legacy text assistant contract	Inactive native path; historical provider adapters remain
POST	/api/{onboarding|sweeps}/{id}/local-voice/{transcribe|speech}	Legacy speech HTTP contracts	Inactive native path; no active sequential STT/TTS chain
POST	/api/sweeps/{id}/local-voice/guidance	Legacy spoken-guidance contract	Inactive native path; current bridge uses Realtime context
WSS	api.openai.com/v1/realtime?model=...	session/audio/transcript/tool/usage events	F01-F03; server credential; independent upstream socket
POST	api.openai.com/v1/responses	Structured scan or search URLs + usage	F05/F07; standard SDK base, configurable; store:false
GET	Discovered public retailer product URL	HTML/text with JSON-LD offers	F07; credential-free retrieval; redirect/address validation
GET	api.frankfurter.dev/v2/rate/{base}/{quote}	JSON pair/date/rate	F07; conversion and appraisal threshold; optional/cacheable
GET	openlibrary.org/search.json	Candidate bibliographic metadata	F05 compatibility branch; not normal current-phone path

Common errors: 401 pairing, 403 Origin, 413 body size, 422 validation; operation-specific alternatives appear in each flow.